

SEEDPRINTS: FINGERPRINTS CAN EVEN TELL WHICH SEED YOUR LARGE LANGUAGE MODEL WAS TRAINED FROM

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ABSTRACT

Fingerprinting Large Language Models (LLMs) is essential for provenance verification and model attribution, yet we argue that current practice measures the wrong thing. Existing fingerprinting methods are evaluated almost exclusively after fine-tuning, where models have already acquired stable signatures from training data, optimization dynamics, or hyperparameters. Because most of a model’s capacity and knowledge are acquired during pretraining rather than downstream fine-tuning, large-scale pretraining is the more fundamental—and more demanding—regime for lineage verification. This paper makes three contributions in that regime. **First**, we expose a fundamental limitation: existing fingerprinting methods become *unreliable* during early pretraining and under distribution shift, because they rely on post-hoc signatures that only emerge after substantial training—violating the classical Galtonian notion of a fingerprint as an intrinsic and persistent identity. **Second**, we introduce a stronger, more intrinsic notion of LLM fingerprinting realized as **SeedPrints**, a method that turns random initialization biases into persistent, seed-dependent identifiers present even before training begins; we show that untrained models exhibit reproducible, seed-induced prediction biases, and that these weak signals remain statistically detectable throughout training to enable high-confidence lineage verification. **Third**, we validate SeedPrints end to end: unlike prior techniques that fail during early pretraining or degrade under distribution shifts, it remains effective across every training stage—from initialization, through large-scale pretraining, to downstream adaptation. Experiments on LLaMA-style and Qwen-style models demonstrate seed-level distinguishability and enable birth-to-lifecycle identity verification, while evaluations on large-scale pretraining trajectories and real-world fingerprinting benchmarks confirm robustness under prolonged training, domain shifts, and parameter modifications. Taken together, these contributions establish that initialization itself imprints a unique and persistent identity on LLMs, forming a true “Galtonian” fingerprint. Code will be released upon acceptance.

1 INTRODUCTION

LLM fingerprints have recently emerged as a tool to identify, attribute, and trace LLMs by examining their observable behaviors (Pasquini et al., 2024; Xu et al., 2024; Yoon et al., 2025; Zhang et al., 2024; Zeng et al., 2024). Their promise is concrete: to give model owners a verifiable link between a suspicious model and its putative original, and thereby detect model theft or unauthorized reuse (Yoon et al., 2025; Zhang, 2025). In this paper we ask whether today’s fingerprints actually deliver on that promise at the stage where it matters most—pretraining—and, finding that they do not, we build one that does.

Much of this literature explicitly borrows the metaphor of biological fingerprints from Francis Galton’s *Finger Prints* (1892) (Galton, 1892):

“A fingerprint is the pattern formed by friction-ridge skin on the fingertips; this ridge configuration is individually unique and essentially permanent across an individual’s lifetime.”

SeedPrints vs Baselines: OLMo-2-7B Pre-training Checkpoints

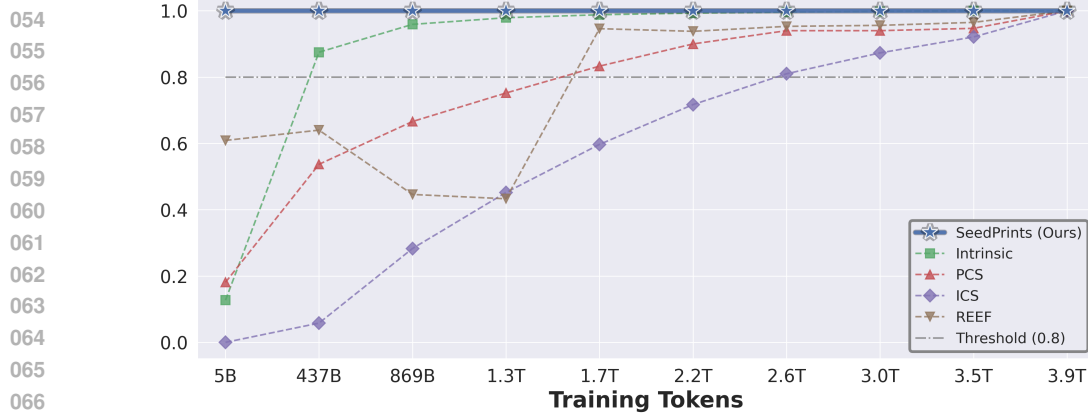


Figure 1: **Motivating our first contribution: existing fingerprinting methods fail to detect model lineage during early pre-training.** We compare five methods on OLMo-2-7B checkpoints spanning 5B to 3.9T training tokens, each tested against the final checkpoint. The y-axis shows the similarity score (higher indicates a stronger lineage signal); the dashed line marks the 0.8 detection threshold. The contrast is stark: every baseline degrades and falls below the threshold at early checkpoints, whereas **SeedPrints** achieves perfect detection ($p \ll 0.001$, plotted as $1 - p$) from the very first checkpoint onward.

The analogy sets a clear bar: an effective fingerprint should be both unique and persistent, present from the very moment of a model’s “birth” at initialization. Our first observation is that most existing so-called fingerprinting approaches fall short of this bar (Pasquini et al., 2024; Xu et al., 2024; Yoon et al., 2025; Zhang et al., 2024; Zeng et al., 2024; Zhang, 2025; Luan et al., 2025; Tsai et al., 2025; Alhazbi et al., 2025). As Figure 1 shows, they fail to reliably detect lineage during early pretraining. The reason is structural: these methods are defined only after models have fully converged, e.g., by extracting patterns from parameters or generated text. Consequently, the separability they achieve is not a birthmark of the model itself but an imprint of surface-level training components—data signatures, optimization dynamics, or hyperparameters. This dependence is also brittle: in Section 5.1, existing baselines often fail to identify lineage once the training data distribution shifts substantially. Such methods therefore function as post hoc identifiers rather than as **Galtonian fingerprints**—the innate, immutable marks that accompany a model from its very beginning. This gap is precisely what our work sets out to close.

In this work, we propose a stricter notion of LLM fingerprinting: an intrinsic property present at *initialization* and detectable at *any* later point during training. Concretely, this paper makes four contributions, which we preview here and return to throughout the paper:

- **Contribution 1 — We expose a fundamental limitation of existing methods.** We show that prior fingerprinting approaches fail to reliably detect lineage during early pretraining and can be easily misled by substantial shifts in the training data distribution, revealing that their apparent success reflects post-hoc training artifacts rather than intrinsic identity. (Section 5.1)
- **Contribution 2 — We discover and operationalize initialization-driven fingerprints.** We uncover that untrained models exhibit seed-dependent biases in their internal representations, forming a weak but persistent identity signal present from initialization (Section 3). Building on this discovery, we introduce **SeedPrints**, a method that isolates these initialization-born fingerprints and remains robust to confounding factors such as data distribution, training duration, and optimization dynamics (Section 4).
- **Contribution 3 — We establish birth-to-lifecycle verifiability.** We demonstrate that SeedPrints distinguishes models differing only in initialization seed—even under identical training pipelines and data orders—and that the fingerprint persists under continual training across diverse datasets. (Section 5.1)
- **Contribution 4 — We validate strong performance in realistic settings.** SeedPrints reliably detects lineage throughout large-scale pretraining trajectories (e.g., OLMo-2-7B Stage 1) without early-stage failure. It also performs strongly under large-scale fine-tuning (up to 700B tokens) and in practical deployment scenarios on the *LeaFBench* benchmark (Shao et al., 2025), which spans 65 models, 7 mainstream model families, and 6 transformations (instruction tuning, fine-tuning, PEFT,

quantization, model merging, and distillation). Here it matches the strongest baseline (near-perfect) while substantially outperforming all others. (Sections 5.2 and 5.3)

2 RELATED WORK

We position our contributions against two families of LLM protection: (i) *watermarking / active fingerprinting* methods that *insert* an identifiable signature into a model or its outputs (Xu et al., 2024; Nasery et al.; Tsai et al., 2025; Sander et al., 2024), and (ii) *passive fingerprinting* methods that *extract* a signature from a model’s pre-existing behaviors without modifying it (Yoon et al., 2025; Zhang et al., 2024; Alhazbi et al., 2025; Pasquini et al., 2024; Suzuki et al., 2025; Zhang, 2025). Our method belongs to the second family, but—as we make precise below—departs from all prior work by grounding the signature in initialization rather than in post-training behavior.

Watermarking and Active Fingerprinting. Active approaches deliberately implant verifiable identifiers for later ownership checks. Classic techniques include backdoor attacks (Adi et al., 2018; Li et al., 2019; Zhang et al., 2018), digital signatures, and hash functions (Guo & Potkonjak, 2018; Zhu et al., 2020). For language models, two forms dominate: *text watermarks*, which bias generation or insert predefined patterns to encode hidden information (Kirchenbauer et al., 2023; Xu et al., 2024; Nasery et al.); and *model weight watermarks*, which embed identifiers into parameters or link them to secret triggers through fine-tuning (Luan et al., 2025; Li et al., 2023). Backdoor-style fingerprints are relatively straightforward and may persist after moderate fine-tuning (e.g., Dasgupta et al. 2024). Their key drawback, and the reason they do not address our target problem, is that these invasive schemes require control of the training process—making them unsuitable for retroactively marking third-party models.

Passive LLM Fingerprinting. In contrast, passive fingerprinting identifies models by analyzing their intrinsic properties without any modification, and techniques vary by model access. With **white-box access**, signatures are extracted from model weights, leveraging intrinsic properties like the distribution of attention matrices (Yoon et al., 2025), the kernel alignment of internal representations (Zhang et al., 2024), or the stable direction of parameter vectors. In the **black-box setting**, fingerprinting instead analyzes input-output behavior, using crafted queries (Pasquini et al., 2024), unique prompt-response pairs (Tsai et al., 2025), stylometry (Alhazbi et al., 2025), or iterative prompting–response games (Iourovitski et al., 2024), though such signals can be less robust to fine-tuning. The common thread—and the limitation our first contribution formalizes—is that all of these methods define their signatures *post hoc*: they read off emergent properties of a completed training process rather than the innate, “Galtonian” fingerprints present from random initialization that our work discovers. As a direct consequence, *they are not designed for lineage verification during large-scale pretraining*, where such post-training signals have not yet emerged. SeedPrints is, to our knowledge, the first passive fingerprint defined at and traceable from initialization.

3 BIASES ORIGINATING FROM INITIALIZATION PERSIST AFTER TRAINING

This section establishes the empirical foundation of our second contribution: the observation that language models exhibit strong prediction biases originating from the random initialization seed, and that these biases remain detectable even after training. This single observation is what makes an initialization-born fingerprint possible; the algorithm in Section 4 then turns it into a usable verification test. We build the observation in two steps—first that initialization creates a seed-specific bias, then that training does not erase it.

Observation 1: initialization induces seed-specific bias profiles. We evaluate a LLaMA-2–style model initialized with seed 123 on 10,000 random input sequences of length 1,024, where tokens are sampled uniformly from the vocabulary. In Figure 2 (Left), we plot the frequency with which each output token (upper, from the final logits) and each hidden dimension (lower, from the last-layer representations) attains the minimum value across the 10,000 random trials. The pronounced non-uniform shape in both plots indicates the presence of extreme output preference bias patterns. Repeating the experiment across different initialization seeds shows that while the overall magnitude of bias remains stable, the specific argmin dimensions depend on the seed (Figure 2, Lower Right). These observations are consistent with recent findings that biases in randomly initialized models arise from inter-sequence representation contraction, driven by asymmetric nonlinear activations in MLP blocks and further amplified by self-attention (Li et al., 2026). Under this view, different initialization seeds induce different contraction directions, resulting in distinct argmin patterns across

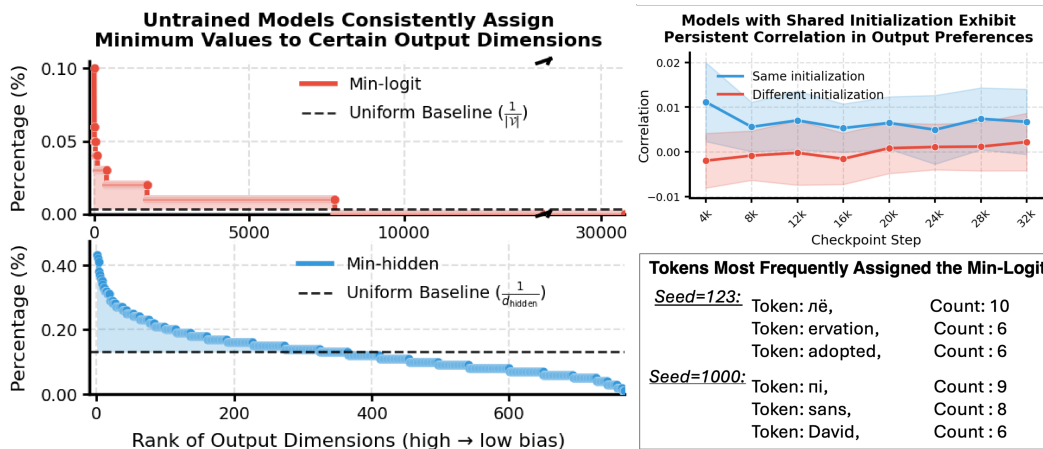


Figure 2: **The core evidence for our fingerprint: initialization-born output bias persists through training.** The three panels correspond to the three ingredients of an initialization fingerprint—bias exists, bias is seed-specific, and bias survives training. *Left (bias exists):* Given completely random inputs, the outputs of a randomly initialized LLaMA-2-style model are far from uniform, and instead exhibit clear bias: certain dimensions are disfavored by the model (i.e., they frequently receive the minimum value across random inputs). This extreme bias appears both in the logits (top, red) and in the final hidden representations (bottom, blue). The dashed line shows the expected frequency under a uniform distribution, and the arrows in the top panel indicate a broken x-axis that omits low-frequency tail ranks. *Upper Right (bias survives training):* During training, models remain weakly correlated in their output bias, across inputs, with the base model they are trained from. This correlation distribution is consistently shifted upward relative to correlations computed with independently initialized base models, showing that initialization-born bias leaves a measurable signal that persists through training. *Lower Right (bias is seed-specific):* The specific dimensions that exhibit strong bias depend on the initialization seed; we illustrate this using the tokens that most frequently receive the minimum logit.

output dimensions. In short, the seed leaves a distinctive, structurally-determined signature the moment the model is born.

Observation 2: training preserves the initialization-born bias. A fingerprint is only useful if it survives; the crux of our contribution is thus that this bias persists. Although training substantially changes output magnitudes, the relative preference over output dimensions induced at initialization is not entirely lost. We train the previously initialized model (seed 123) for one epoch on OpenWeb-Text (Gokaslan et al., 2019) and evaluate intermediate checkpoints. We focus on the $m = 50$ output dimensions that are most frequently assigned minimum values by the initialized model. For each checkpoint, we compute the correlation between its responses and those of the initialized model across random inputs, restricted to these dimensions. Intuitively, this measures how similarly the checkpoint and the initialized model rank or disfavor these dimensions across inputs. As a baseline, we compute the same correlation with other independent initialized models (seed 1000). As shown in Figure 2 (Upper Right), although the absolute correlation values are small, correlations for models from the same initialization are consistently shifted upward relative to the independent baseline, and stabilize rather than decaying to zero. Together, Observations 1 and 2 deliver the takeaway that motivates the rest of the paper: initialization leaves a weak but statistically detectable bias signal that persists throughout training—exactly the “born-and-persistent” property a Galtonian fingerprint requires.

4 ALGORITHM

This section delivers the methodological half of our second contribution: it turns the persistence observation into a concrete, threshold-free verification test. Section 3 showed that models from the same initialization lineage retain weak but consistent agreement on a subset of output dimensions (e.g., those most disfavored by the untrained model under random inputs). The design goal follows directly: given any two models—and *without* access to the true initialized model—uncover the dimensions that carry this bias signal, and test whether their agreement is statistically significant.

The overall procedure is summarized in Algorithm 1; we describe each of its three components (identity-dimension extraction, correlation statistics, and hypothesis testing) below.

Extract identity dimensions between any two models. Let $X = \{x_i\}_{i=1}^n$, where each $x_i \in \mathbb{R}^{\ell \times d}$ denotes a random sequence of length ℓ . Each random input x_i can be instantiated either by uniformly sampling ℓ token IDs from the vocabulary, or by directly sampling ℓ independent vectors from a d -dimensional isotropic Gaussian distribution. For any model g , we define its mean response vector as $\bar{g} := \frac{1}{n} \sum_{i=1}^n g(x_i) \in \mathbb{R}^{d_{\text{out}}}$, where d_{out} denotes the output dimensionality. Here, $g(x_i)$ can denote either the model’s logits or its final-layer representations. We then identify its disfavored dimensions as the set of bottom- m coordinates of $\bar{g}$:

$$\mathcal{M}_g := \arg \min_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} \bar{g}_j.$$

We deliberately use the mean response vector to extract bias signals: it is a more stable, noise-robust alternative to per-sample argmin frequency (counting how often each dimension attains the minimum value, as illustrated in Figure 2), and it also performs well empirically. This choice is not merely convenient—we prove its stability advantage formally in Appendix C.1.

Given two models f and f' , we define their *identity dimensions* as the intersection $\mathcal{S} := \mathcal{M}_f \cap \mathcal{M}_{f'}$. The intuition is simple: two models sharing the same initialization lineage should independently flag a similar set of disfavored dimensions, producing non-trivial overlap in their bottom- m sets. The intersection therefore acts as a built-in mutual verification step—it suppresses spurious correlations from architectural similarity or shared training dynamics and isolates the bias signals most likely to originate from shared initialization lineage. When one model is closer to initialization, its disfavored dimensions can also serve directly as an anchor.

Correlation statistics. Restricting both models to the identity dimensions $\mathcal{S}$, we compare their responses across random inputs.¹ For each $j \in \mathcal{S}$, we compute a Kendall–Tau rank correlation:

$$\tau_j = \text{KendallTau}(\{f(x_i)_j\}_{i=1}^n, \{f'(x_i)_j\}_{i=1}^n).$$

This yields a set of correlation statistics $T = \{\tau_j\}_{j \in \mathcal{S}}$ that captures the distribution of agreement between the two models across identity dimensions. Consistent with Section 3, same-lineage models exhibit distributionally higher agreement on these dimensions than unrelated models—the signal our test is designed to detect.

Hypothesis testing via null distribution. Because the correlation distributions for same-lineage and different-lineage models can overlap, thresholding individual statistics is insufficient; this is exactly the pitfall that makes similarity-threshold baselines unreliable. We instead test whether the observed correlations are significantly larger than expected under the null hypothesis of no lineage relation. Ideally, this null should be estimated from the correlation distribution between independent models as in Figure 2. However, obtaining such a baseline empirically requires comparisons across many independently initialized models, which is computationally expensive. As shown in Appendix B, the null correlation distribution between independently initialized models is well approximated by a Gaussian distribution. This naturally motivates a cheaper simulation-based alternative: constructing an empirical null distribution using Gaussian surrogates passed through the same pipeline, as summarized in Algorithm 1.

While this simulation-based null already provides a strong empirical approximation, it still introduces unnecessary randomness and computational overhead: its precision is limited by the number of simulation trials, results can vary across runs, and comparing two empirical distributions typically requires additional assumptions from the adopted test, such as the one-sided t-test or the Mann–Whitney U test. Since the null distribution of Kendall–Tau (under weak independence) is available in closed form, we further replace this simulation-based procedure with an analytical test, while also providing both empirical and analytical implementations in our code repository. Specifically, under the null hypothesis, each Kendall–Tau correlation is approximately zero-mean with known variance σ^2 . By the central limit theorem, as the number of identity dimensions $|\mathcal{S}|$ increases,

¹Before computing the correlations, we apply a row-wise softmax normalization to each sample’s output vector. This removes scale differences across different models and dimensions while preserving the relative ordering that carries identity information for rank-based statistics such as Kendall–Tau. A relatively high temperature is used to avoid degenerate one-hot behavior.

Algorithm 1 Distribution Correlation Test on Identity Dimensions

Require: base model f , suspicious model f' ; random inputs $X = \{x_i\}_{i=1}^n$; fingerprint size m ; significance level α

► **Step 1: Localize biased dimensions**

- 1: Compute average outputs $\bar{f}, \bar{f}'$ over X
- 2: $\mathcal{M}(f), \mathcal{M}(f') \leftarrow$ bottom- m dimensions of $\bar{f}, \bar{f}'$
- 3: $\mathcal{S} \leftarrow \mathcal{M}(f) \cap \mathcal{M}(f')$ (identity dimensions)

► **Step 2: Form correlation distribution**

- 4: Normalize $f(x_i)_{\mathcal{S}}, f'(x_i)_{\mathcal{S}}$ for each $x_i \in X$ across identity dimension
- 5: **for** each $s_j \in \mathcal{S}$ **do**
- 6: $\tau_j \leftarrow \text{KendallTau}(\{f(x_i)_{s_j}\}_{i=1}^n, \{f'(x_i)_{s_j}\}_{i=1}^n)$ (correlation in input-wise preferences)
- 7: **end for**
- 8: $\mathcal{T} \leftarrow \{\tau_j\}_{j=1}^{|\mathcal{S}|}$

► **Step 3: Hypothesis test against null**

- 9: Construct $\mathcal{T}_{\text{null}}$ by applying the same pipeline to two Gaussian matrices $Y^{(1)}, Y^{(2)} \sim \mathcal{N}(0, I)^{N \times d_{\text{out}}}$
- 10: Test $H_0 : \mathcal{T} = \mathcal{T}_{\text{null}}$ vs. $H_1 : \mathcal{T} > \mathcal{T}_{\text{null}}$
- 11: **Return** SameLineage $\leftarrow \mathbf{1}(p\text{-value} < \alpha)$

the distribution of $\bar{\tau} = \frac{1}{|\mathcal{S}|} \sum_{j \in \mathcal{S}} \tau_j$ is well approximated by $\bar{\tau} \sim \mathcal{N}\left(0, \frac{\sigma^2}{|\mathcal{S}|}\right)$. While softmax normalization introduces weak dependencies across dimensions, these effects are mitigated by the use of a relatively high temperature ($T = 10$), making the independence assumption a reasonable approximation in practice. We empirically validate that this Gaussian approximation closely matches the distribution obtained from the full pipeline (Appendix B). We therefore compute the one-sided z-score $z = \frac{\bar{\tau}}{\sigma/\sqrt{|\mathcal{S}|}}$, and the corresponding p-value $p = 1 - \Phi(z)$, where Φ is the standard normal cumulative distribution function. We reject the null hypothesis and declare shared lineage if $p < 0.01$.

Practical instantiation. Model outputs can be taken as either logits or hidden representations. Both are consistent with our formulation, but using logits requires substantially more random inputs to reliably estimate bias patterns as the vocabulary size grows, incurring heavy computational overhead (see Appendix C.2 for the theoretical analysis). For efficiency and stability, we therefore adopt final-layer hidden states as the default model output. The payoff of this design—and the property that distinguishes SeedPrints from every similarity-based baseline—is that the resulting p-value *is* the decision: it provides a direct statistical test for lineage without any heuristic similarity threshold.

5 EXPERIMENTS

Our experiments are organized to substantiate Contributions 1, 3, and 4 in turn. Section 5.1 establishes Contribution 3 by showing that SeedPrints functions as a genuine fingerprint: (i) it enables *birth verification at the seed level*, and (ii) it remains *verifiable across the entire training lifecycle*. The same experiments substantiate Contribution 1, exposing that existing baselines cannot verify lineage at early pretraining stages and lean on data-dependent signals that collapse under significant distribution shifts. Sections 5.2 and 5.3 then establish Contribution 4 under realistic large-scale and deployment conditions. Unless otherwise noted, we report results using the analytical null with *hidden state* outputs; complementary results using the empirical null with both *logit* and *hidden state* outputs, evaluated with the one-sided *t*-test and the Mann–Whitney *U* test, appear in Appendix D.2. Concretely, Section 5.3 evaluates all methods under practical infringement scenarios using *LeaFBench* (Shao et al., 2025): across 65 distinct model instances spanning 6 representative post-development techniques, SeedPrints matches the best baselines in post-training settings and stays robust to diverse deployment transformations.

Baselines. To make the comparison fair and comprehensive, we consider four passive fingerprinting baselines (weight- or representation-based), each representative of a distinct signature source. *Intrinsic fingerprint* (Yoon et al., 2025) (or *PDF* in some papers) compares models via the similarity of the layerwise standard-deviation profiles of attention parameters. *REEF* (Zhang et al., 2024) computes centered-kernel-alignment (CKA) similarity between feature representations from the same samples across two models. *PCS* and *ICS* (Zeng et al., 2024) (or collectively as *HuRef* in some papers) are weight-similarity methods: PCS flattens all parameters and measures cosine similarity; ICS forms invariant terms from the weights and measures cosine similarity on those invariants. In Section 5.3, we additionally include a (weaker) gradient-based fingerprint *Gradient* (Shao et al., 2025). Following Zhang et al. (2024), we use a **0.8 similarity threshold** for binary decisions.

Table 1: **Uniqueness at birth** (supports Contribution 3). Trained models keep their initialization fingerprint SeedPrints separates models ($p < 0.01$), whereas all four baselines fail to link a model to its initialized with different seeds: own seed.
every cross-seed p -value stays above 0.01.

Model Pairs	p -value (> 0.01)
s_{42} vs. s_{2000}	0.116
s_{123} vs. s_{42}	1.000
s_{1000} vs. s_{123}	0.327
s_{2000} vs. s_{1000}	1.000

Table 3: **Uniqueness is intrinsic, not data-driven** (supports Contribution 3). Identical corpus and training order do not make fingerprints converge across different initializations ($p > 0.01$).

Model Pair	p -value (> 0.01)
s_{42}^{init} vs. s_{123}^{base}	0.573
s_{123}^{init} vs. s_{2000}^{base}	0.724
s_{2000}^{init} vs. s_{1000}^{base}	0.109
s_{1000}^{init} vs. s_{42}^{base}	0.883

Model Pairs	SeedPrints p -value	Baselines			
		Intrinsic	REEF	PCS	ICS
s_{42}^{init} vs. s_{42}^{base}	3.37e-26✓	-0.021×	0.375×	0.580×	0.196×
s_{123}^{init} vs. s_{123}^{base}	3.16e-33✓	0.149×	0.369×	0.581×	0.188×
s_{1000}^{init} vs. s_{1000}^{base}	1.12e-21✓	-0.252×	0.381×	0.581×	0.188×
s_{2000}^{init} vs. s_{2000}^{base}	3.70e-31✓	-0.337×	0.331×	0.581×	0.188×

Table 4: **Survival under domain shift (the decisive test for Contribution 1)**. Fingerprint persistence under continual training on diverse datasets (base model: seed 1000, corpus openwebtext); baselines misclassify true descendants once the corpus shifts to code, while SeedPrints attributes lineage correctly.

Setting	SeedPrints p -value	Baselines			
		Intrinsic	REEF	PCS	ICS
Continual corpus (seed)					
TinyStories (1000)	≈ 0 ✓	1.000✓	0.759×	0.999✓	0.996✓
TinyStories (123)	1.000✓	0.950×	0.658✓	0.332✓	0.012✓
the_stack (1000)	≈ 0 ✓	0.489×	0.557×	0.585×	0.123×
the_stack (123)	1.000✓	0.445✓	0.580✓	0.301✓	0.026✓

Note, in all experiment tables, cell colors indicate lineage: with green denotes models from the same source, and red denotes different sources. For example, s_{42}^{init} vs. s_{42}^{base} compares a model initialized with seed 42 and its continued-pretrained counterpart, hence green. By contrast, s_{2000}^{init} vs. s_{42}^{base} compares a seed-2000 initialization with a model trained from a seed-42 initialization, hence red. Additionally, ✓ denotes a correct detection, while × denotes an error.

5.1 BIRTH-TO-LIFECYCLE “BIOMETRIC” FINGERPRINTING

To establish Contribution 3, we test the two properties a biometric fingerprint must satisfy—uniqueness at birth and persistence through life—one at a time. We train 12-layer, 12-head LLaMA-style models (Touvron et al., 2023) with RoPE (Su et al., 2021) and Qwen-style models (Team, 2024) from scratch. The main paper reports LLaMA-style models and defers Qwen-style models to Appendix D; the conclusions are consistent across the two.

Finding 1: different initialization seeds produce distinct fingerprints (uniqueness at birth). Table 1 reports p -values from our correlation tests between pairs of models initialized with different random seeds (42, 123, 1000, and 2000). Every p -value is > 0.01 , so our method reliably separates models born from different seeds. Distinct seeds thus yield distinct fingerprint behaviors, letting us tell models apart “at birth.”

Finding 2: training preserves the initialization fingerprint (persistence). Table 2 compares each initialization model s_i^{init} with its descendant s_i^{base} trained on the OpenWebText dataset (Gokaslan et al., 2019) ($\approx 10B$ tokens). Every seed–model pair yields $p < 0.01$: the bias profiles stay strongly correlated and the pair is correctly identified as one lineage. In short, the trained model inherits the same fingerprint as its initialization. The same table also delivers direct evidence for Contribution 1: the four baselines, without exception, fail to distinguish across seeds, revealing that their separability stems from training-induced artifacts rather than from initialization.

Finding 3: identical data and order do not make fingerprints converge (uniqueness is intrinsic, not data-driven). In Table 3, all four “suspicious” models s_i^{base} for $i \in \{42, 123, 100, 2000\}$ are trained on exactly the same corpus (OpenWebText) and in the same data order (we fix the training seed to lock the data order); the only difference is the initialization seed i . Across all cross-seed pairs, p -values remain > 0.01 , in sharp contrast to the near-zero values in Table 2. The fingerprint therefore

remains seed-specific even under identical data and curriculum—confirming that the signal comes from initialization, not from what the model was trained on.

Finding 4: continual training on diverse datasets does not confound the fingerprint (the decisive test for Contribution 1). The experiments so far demonstrate that SeedPrints can act as a biometric fingerprint. We stress that the inability of prior work to distinguish models with different seeds is not, by itself, a copyright weakness, since initialization seeds have no clear copyright status. The genuinely damaging failure mode is different: prior attribution can be misled by *data distribution*, *failing to recognize lineage once the training distribution shifts substantially during continued training*. This finding is what makes Contribution 1 more than a technicality.

To probe this, in Table 4 we continue training a base model (seed 1000, pretrained on OpenWebText (Gokaslan et al., 2019)) on two very different datasets: TinyStories (Eldan & Li, 2023) (synthetic children’s stories) and The Stack (Kocetkov et al., 2022) (permissively licensed GitHub code). We compare (i) true descendants trained from the base against (ii) *distractors* derived from a different base model (seed 123). Because baselines are insensitive to initialization differences, the distractor base is pretrained with a different data order on OpenWebText and then continually trained on the *same* target corpus to form the distractor. The question is which descendant an attribution method identifies as truly sharing lineage with the base.

The results are decisive. Prior baselines all fail under the code setting (The Stack), misclassifying true descendants as distractors—a clear sign that they track domain similarity rather than lineage identity: TinyStories stays close in distribution to the pretraining corpus (OpenWebText), whereas The Stack diverges sharply, and this large shift is enough to bypass detection. SeedPrints, by contrast, correctly attributes lineage across both corpora. Our fingerprint is therefore not a proxy for data distribution: it survives substantial domain shift and persists well beyond the initial pretraining stage.

5.2 ALL-STAGE VERIFIABLE FINGERPRINTS

This subsection delivers the first half of Contribution 4: that SeedPrints is verifiable at *every* stage of training, not just after convergence. We build the case from a controlled setting up to realistic large-scale runs.

Controlled pretraining trajectory. Can our method reliably identify its offspring along the training trajectory? We first answer this in a controlled pretraining experiment on OpenWebText, where the initialization seed is known, by testing lineage at intermediate checkpoints. Detection is consistent across all stages (Appendix D.2.1, Figure 5).

Large-scale pretraining. We then raise the difficulty to realistic foundation-model training, analyzing ten checkpoints from the OLMo-2-7B (OLMo et al., 2024) Stage-1 run (5B $\rightarrow$ 3.9T tokens): the final checkpoint is the target model and all earlier checkpoints are the tested ancestors. Unlike the controlled OpenWebText setting, this stage spans heterogeneous corpora, longer optimization, and large-scale representation drift. We plot $1 - p$ for directional consistency with similarity-based baselines; all p -values are $\ll 0.01$ and numerically close to zero.

Figure 1 makes the headline result visible: **all existing baselines fail to verify lineage during early pretraining** (e.g., $ICS \approx 0$, $PCS < 0.3$, and $REEF < 0.6$ within the first **trillion** tokens), whereas SeedPrints stays verifiable throughout and only strengthens with training. This exposes a critical gap in current evaluation practice—the crux of Contribution 1 at scale: **most prior work verifies lineage only after fine-tuning, where detection is substantially easier**. Because large-scale pretraining itself reshapes representations and strengthens late-stage lineage signals, evaluating only at convergence gives a *false sense of safety*. We argue that lineage verification must prioritize the early pretraining stages, where misuse is hardest to detect, rather than relying on late-stage checkpoints alone.

Large-scale finetuning. The second half of Contribution 4 concerns the downstream end of the lifecycle. Under standard finetuning-stage evaluation, we test suspect models fine-tuned from Llama-2-7b (base model) with data volumes ranging from 5 million to 700 billion tokens. The suspects span diverse downstream variants: Llama-2-finance-7b (Heenan, 2023), Vicuna-1.5-7b (Chiang et al., 2023), WizardMath-7b (Luo et al., 2023), Chinese-LLaMA-2-7b (Chen et al., 2023), Code-Llama-7b (Meta AI, 2024), and Llemma-7b (Azerbayev et al., 2023), with fine-tuning data volumes of 5M, 370M, 1.8B, 13B, 500B, and 700B tokens, respectively. As Table 5 shows, SeedPrints maintains $p < 0.01$ across every setting, whereas the weight-based baselines PCS and ICS break down at the highest token counts.

Table 5: **Downstream half of Contribution 4: verifiability under large-scale finetuning.** Each row compares a target model against LLaMA-2-7B. **SeedPrints** reports the p -value from our correlation test (< 0.01 indicates a strong signal) and holds across all six variants up to 700B tokens, whereas the weight-based baselines PCS and ICS fail at the highest token counts. The four baselines all report similarity scores (threshold = 0.8, higher = better).

Model	# Tokens	SeedPrints p	Intrinsic $\uparrow$	REEF ($\uparrow$)	PCS ($\uparrow$)	ICS ($\uparrow$)
Llama-2-finance-7B (Heenan, 2023)	5M	10^{-41955} ✓	1.0000 ✓	0.9950 ✓	0.9979 ✓	0.9952 ✓
Vicuna-1.5-7B (Chiang et al., 2023)	370M	$10^{-103043}$ ✓	1.0000 ✓	0.9985 ✓	0.9985 ✓	0.9949 ✓
Wizardmath-7B (Luo et al., 2023)	1.8B	$10^{-180550}$ ✓	1.0000 ✓	0.9979 ✓	1.0000 ✓	0.9994 ✓
Meditron-7B (Chen et al., 2023)	48B	10^{-42566} ✓	0.9990 ✓	0.9978 ✓	1.0000 ✓	0.9817 ✓
CodeLlama-7B (Meta AI, 2024)	500B	10^{-3552} ✓	0.9480 ✓	0.9947 ✓	0.6863 ✗	0.3369 ✗
Llemma-7B (Azerbayev et al., 2023)	700B	10^{-5136} ✓	0.9470 ✗	0.9984 ✓	0.6682 ✗	0.2905 ✗

5.3 ROBUSTNESS UNDER REALISTIC DEPLOYMENTS

The final piece of Contribution 4 is robustness under real deployment. Real-world deployments routinely apply *parameter-altering* adaptations to foundation models (e.g., fine-tuning, PEFT, quantization) that can weaken or distort fingerprint signals, and SeedPrints is designed to survive this regime. To test that claim at scale, we adopt *LeaFBench* (Shao et al., 2025), a benchmark for language-model copyright auditing.

Across Source Models. We consider two source-model types for auditing: *Pre-Trained (PT)* and *Instruction-Tuned (IT)*. For each, the task is to separate *derivative* models (produced by post-training adaptation) from *independent* models (trained without the source weights)—mirroring the common auditing scenario of testing a lineage claim for either a PT base or its IT variant. The evaluation covers 65 models in total; the full list of model names and HuggingFace repositories is in Appendix E.

Metrics. We evaluate model detection performance using (i) the Area Under the ROC Curve (AUC) and (ii) the Kolmogorov–Smirnov (KS) statistic (Berger & Zhou, 2014), which measures the maximum separation between score distributions of derivative and independent models.

One caveat matters for interpreting these metrics, and it actually understates our advantage. SeedPrints produces per-model p -values from statistical tests, which differ fundamentally from the similarity scores used by the baselines: similarity scores are linear measures whose differences carry proportional meaning, whereas p -values are tail probabilities indicating the rarity of an observation under the null and cannot be read on a linear scale. To compare against baselines that compute AUC from similarity scores, we convert our p -values into scores via $s = 1 - p$. This conversion is not fully faithful to the meaning of p -values and is, if anything, unfavorable to us—small p -values change on an exponential scale while threshold sweeping is linear, blurring fine-grained differences—yet SeedPrints still attains comparable or superior performance. For evaluation we therefore (i) compute AUC for all methods by sweeping thresholds over the scores s and (ii) report the KS statistic as a threshold-free measure of distributional separability. This protocol keeps us compatible with baseline pipelines while preserving our statistical interpretability. We omit Manhattan distance, as absolute distances between p -values are not meaningful.

This distinction is itself a practical contribution, not a footnote. In most real scenarios, calibrating an optimal similarity threshold is infeasible; we report AUC only for comparability. **Given any pair of models, SeedPrints outputs a p -value that yields a definitive same-lineage decision without any tunable threshold**, which is what makes it reliable in deployment where baselines have no principled cutoff.

Parameter-Altering Techniques. To assess fingerprinting robustness, we evaluate models produced via (1) Instruction tuning (Instruct), (2) General-purpose fine-tuning (Finetune), (3) Parameter-efficient fine-tuning (PEFT), (4) Quantization, (5) Model merging (Merge), and (6) Distillation.

5.3.1 FAMILY-WISE EFFECTIVENESS

There are seven model families in total. For each family, we treat both its pretrained base model (PT) and its instruction-tuned model (IT) as separate source models and compare them against all other models in the pool of 65, reporting results in Table 6. We omit the TinyLlama-1.1B families from Table 6, as they are trivial for all methods. Each column corresponds to one family (AUC

Table 6: **Family-wise evidence for Contribution 4.** Performance comparison of LLM fingerprinting methods across source models; SeedPrints ties the strongest post-training baselines (Intrinsic, ICS) at overall AUC 0.992. “PT” and “IT” denote pre-trained and instruction-tuned source models. The per-family breakdown covers the 6 families that have both PT and IT models. The “Overall” scores are computed over all test model pairs, not averaged across families.

Method	Metric	Qwen2.5-7B		Qwen2.5-14B		Llama3.1-8B		Mistral-7B		Gemma2-2B		Llama2-7B		Overall
		PT	IT	PT	IT	PT	IT	PT	IT	PT	IT	PT	IT	
REEF	AUC ↑	0.796	0.862	0.947	0.913	1.000	0.999	0.997	0.997	0.963	0.968	0.750	0.706	0.915
	KS Statistic ↑	0.492	0.633	0.875	0.735	0.987	0.987	0.975	0.980	0.950	0.837	0.750	0.694	0.739
Gradient	AUC ↑	0.657	0.743	0.763	0.789	0.897	0.904	0.872	0.682	1.000	1.000	0.650	0.768	0.801
	KS Statistic ↑	0.354	0.423	0.500	0.563	0.646	0.705	0.725	0.500	1.000	1.000	0.275	0.479	0.508
ICS	AUC ↑	1.000	1.000	0.997	0.997	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.994
	KS Statistic ↑	1.000	1.000	0.975	0.979	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.943
Intrinsic	AUC ↑	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.994
	KS Statistic ↑	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.989
SeedPrints	AUC ↑	0.992	0.994	0.988	0.990	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.992
	KS Statistic ↑	0.985	0.987	0.975	0.980	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.986

Table 7: **Alteration-wise robustness for Contribution 4.** Performance across parameter-altering techniques; SeedPrints stays > 0.9 on all six, with its only dip on *Merge*, keeping it in a virtual tie with the best post-training baselines.

Method	Metric	Instruct	Finetune	PEFT	Quantization	Merge	Distillation	Overall Avg
REEF	AUC ↑	0.996	0.872	0.994	0.871	0.954	0.858	0.924
	KS Statistic ↑	0.949	0.637	0.940	0.812	0.851	0.615	0.801
Gradient	AUC ↑	0.895	0.729	0.776	0.756	0.832	0.887	0.812
	KS Statistic ↑	0.756	0.441	0.469	0.586	0.519	0.679	0.575
ICS	AUC ↑	1.000	0.987	1.000	1.000	0.981	1.000	0.995
	KS Statistic ↑	1.000	0.925	1.000	1.000	0.941	1.000	0.978
Intrinsic	AUC ↑	1.000	0.980	1.000	1.000	1.000	1.000	0.997
	KS Statistic ↑	1.000	0.971	1.000	1.000	1.000	1.000	0.995
SeedPrints	AUC ↑	1.000	0.995	0.990	1.000	0.959	1.000	0.991
	KS Statistic ↑	1.000	0.990	0.980	1.000	0.941	1.000	0.985

computed over the corresponding tests with a family-calibrated threshold), whereas the overall score is computed across all test pairs with a global threshold; the overall score is therefore *not* the average of the per-family scores. The headline finding is a near-tie at the top: *Intrinsic*, *ICS*, and our *SeedPrints* are all essentially saturated across every alteration (overall AUC 0.994, 0.994, and 0.992; KS 0.989, 0.943, and 0.986), with family-wise scores at the full mark in most cases, while the remaining baselines *REEF* and *Gradient* lag substantially. The takeaway reinforces Contribution 4: although SeedPrints is designed to capture intrinsic seed-level fingerprints, it also functions as a biometric-like fingerprint that tracks identity reliably and persistently across diverse model families—matching the strongest post-training baselines on their own turf.

5.3.2 ALTERATION-WISE ROBUSTNESS

Breaking the results down by alteration type sharpens the same message. *Instruct* is consistently the easiest case (even the weakest baseline, Gradient, reaches AUC 0.895), whereas *Finetune* and *Merge* are the hardest for every method—no method achieves perfect *Finetune* performance. *PEFT*, *Quantization*, and *Distillation* are more mixed: REEF drops notably under *Quantization* (0.996 → 0.871) and *Distillation* (0.996 → 0.858) yet stays resilient to *PEFT* (0.994), while *PEFT* and *Quantization* are the tougher cases for Gradient (0.895 → 0.776/0.756). The three stronger methods stay robust to all transformations (all scores > 0.9). For SeedPrints, the only notable dip is *Merge*, where it drops to 0.959 by missing a single merge model out of seventeen; we report this honestly, as it indicates that weight-space interpolation with unrelated weights can dilute the initialization signal below the detection threshold. Averaged across all six alteration types, Intrinsic edges out the highest overall AUC, followed closely by ICS and SeedPrints—placing our method in a virtual tie with the best post-training baselines even where their signals are strongest, while it alone remains verifiable during pretraining (Contribution 4).

5.4 ABLATION STUDY ON HYPERPARAMETERS

Beyond the headline contributions, we show that SeedPrints is easy to configure. The method has three hyperparameters: the number of random input sequences n , the length of each random sequence l , and the fingerprint size m . The first two control the overall size of the random inputs, while m specifies how many low-mean coordinates enter the test. Throughout the main experiments in Sections 5.2 and 5.3 we set $n=2000$, $l=1024$, and $m=0.1d_{\text{out}}$, where d_{out} is the hidden-state dimensionality, so the method scales to different open-source models. Here we vary each hyperparameter while fixing the others to their defaults ($n = 2000$, $m = 400$, $l = 768$) and report empirical accuracy and false-positive rates at a fixed significance level $\alpha = 0.01$ in Table 8.

Table 8: Ablation study under significance level $\alpha = 0.01$.

Two trends emerge, and both match our theory. Increasing n and l consistently improves identification, exactly as predicted by the analysis in Section C.2: accurate recovery of the population-level bias requires $n = \Omega\left(\frac{\log d_{\text{out}}}{\gamma^2}\right)$ queries, where γ is the bias margin separating biased coordinates from the rest. A target dimension therefore needs a minimum sample budget, which is why small n underperforms. Because γ grows with l through accumulation effects, longer sequences strengthen the separation; so under a fixed (insufficient) number of sequences, increasing l lowers the required query count and improves detection.

Ablation on n	200	500	2000
ACC	0.7368	0.7778	0.9375
Empirical FPR	0	0	0
Ablation on l	512	768	1024
ACC	0.9048	0.9375	1.0000
Empirical FPR	0.04	0	0
Ablation on m	200	400	800
ACC	0.9375	0.9375	0.875
Empirical FPR	0.04	0	0.1463

The fingerprint size m behaves non-monotonically, revealing a sweet spot. Too small an m leaves the selected coordinates unstable, with too few empirical observations to form a reliable distribution; too large an m pulls in irrelevant coordinates whose noise shrinks γ and would demand quadratically more samples to compensate. Empirically, m in the 200–400 range works well across model sizes, matching our default heuristic $m \approx 0.1d_{\text{out}}$ for typical $d_{\text{out}} \in [3000, 5000]$.

6 CONCLUSION

This work introduced *SeedPrints*, a stronger, intrinsic notion of LLM fingerprinting that traces a model’s lineage back to its random initialization, and it delivered on four fronts. We **exposed** that prior fingerprints fail during early pretraining and under distribution shift because they read post-hoc training artifacts (Contribution 1); we **discovered** that initialization itself leaves a persistent “Galtonian” fingerprint—untrained models already exhibit stable, seed-dependent token-preference patterns that persist, statistically and measurably, throughout training—and turned this into a distribution-correlation test over *identity dimensions* that detects shared lineage with calibrated significance (Contribution 2); we **established** birth-to-lifecycle verifiability across LLaMA- and Qwen-style families (Contribution 3); and we **validated** robustness across a broad suite of realistic deployments—continued pretraining on divergent corpora, instruction tuning, PEFT, quantization, merging, and distillation (Contribution 4). Together these make SeedPrints a practical, threshold-free tool for provenance and copyright auditing that remains reliable under domain shift.

Limitations and Future Directions We state the scope of our contributions plainly. SeedPrints uncovers fingerprint signals rooted in initialization-stage bias and requires access to model outputs (e.g., logits or hidden states); it may not apply when only restricted API access is available. We view this not as a weakness of the approach but as evidence that the pursuit of *true fingerprints* for LLMs remains at an early stage: most existing approaches (especially black-box ones) rely on signals emerging in later training stages or from alignment-driven artifacts, and therefore cannot robustly protect models at the pretrained stage. This is precisely why reliable all-stage fingerprints—the gap our contributions begin to close—are so urgently needed. See full discussions in Section A.2.

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A LLM USAGE AND LIMITATION DISCUSSIONS

A.1 LLM USAGE

We used AI assistants (ChatGPT and Gemini) only for writing and formatting support: grammar and style checks, clarity improvements to figure and table captions, and other surface-level edits. For programming-related tasks, we occasionally used GitHub Copilot and Claude as coding assistants, e.g., for code auto-completion and debugging hints. None of the paper’s scientific contributions, claims, or results were generated by these tools.

A.2 LIMITATIONS AND FUTURE DIRECTIONS

We delimit the scope of our contributions honestly. Although this work uncovers seed-born, training-persistent biases that serve as fingerprint signals, the method remains within the white-box paradigm: it may not apply when only restricted API access is available, for instance when responses are deterministic or safety-filtered. We emphasize, however, that the pursuit of *true fingerprints* for LLMs is still at a very early stage even under the white-box setting. Most existing fingerprinting approaches rely on training-induced signatures or alignment behaviors and thus cannot robustly protect pretrained or early-stage models. By analogy to human fingerprints, which remain stable regardless of age or environment, we argue that a true LLM fingerprint should persist across continued training and deployment rather than reflect incidental training artifacts—the standard our Galtonian formulation is built to meet. Achieving this under black-box access, where a substantial performance gap to white-box methods remains, is a challenging open direction. We hope this work motivates further research toward principled, robust fingerprinting frameworks for large models.

B VALIDATION OF THE EMPIRICAL AND ANALYTICAL NULL DISTRIBUTION

The threshold-free decision rule that anchors Contribution 2 rests on a valid null distribution; this section supplies the justification. Our hypothesis test in Section 4 decides whether two models share lineage by comparing the distribution of their correlation statistics over random inputs against a null distribution that models the correlation statistics between two *independent models evaluated on random inputs*.

We consider two ways to approximate this null: (i) an empirical simulation-based null obtained by applying the full pipeline to Gaussian surrogate outputs, and (ii) an analytical null derived from the known distribution of Kendall–Tau correlations. We justify both in three steps: (i) characterizing the true null using independently initialized models, (ii) introducing the Gaussian surrogate as a tractable simulation-based proxy, and (iii) validating the analytical approximation.

B.1 EMPIRICAL TRUE NULL FROM INDEPENDENTLY INITIALIZED MODELS

The ideal null corresponds to the correlation statistics between two independent models evaluated on random inputs, and the cleanest way to realize this independence is to compare two independently initialized models (different random seeds, no training). We estimate this empirical null over ~ 2500 such pairs, computing Kendall–Tau correlations on 10,000 random inputs (as in Section 3). Figure 3 shows the result that our test relies on: the true null is empirically a Gaussian centered around zero.

B.2 GAUSSIAN SURROGATE AS A SIMULATION-BASED APPROXIMATION

Estimating the null directly from independently initialized models is computationally expensive, requiring many models and repeated evaluations. For a tractable alternative, we introduce a simulation-based null that uses Gaussian surrogates as proxies for model outputs under the null.

Two considerations justify this choice. First, approximating neural-network outputs with Gaussian distributions is standard in the auditing literature and supported by both empirical and theoretical studies (Carlini et al., 2022; Lee et al., 2019). Second, our pipeline depends only on relative ordering and selection operations (rank correlations and bottom- m selection) and not on semantic structure in the inputs; moreover, randomly initialized models have independent parameter matrices, so their outputs under the null carry no strong structured dependencies. Sampling independent Gaussian

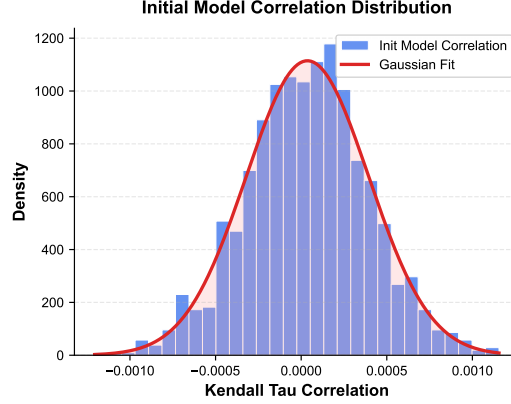


Figure 3: **The null is Gaussian, validating our threshold-free test.** The empirical correlation distribution over ~ 2500 pairs of independently initialized models, evaluated on random inputs, closely matches a Gaussian centered at zero.

matrices is therefore a simple, tractable proxy for the null behavior of independently initialized models.

We construct the surrogate null by sampling Gaussian matrices and applying the full pipeline, including identity-dimension selection and correlation computation:

1. Sample two Gaussian matrices of shape $\mathbb{R}^{N \times d_{\text{out}}}$,
2. Extract the most biased dimensions by selecting the bottom- m ranked average dimensions,
3. Take the intersection set $\mathcal{S}$ and compute correlations on it.

By applying the same selection procedure to Gaussian samples, this construction naturally captures the dependency patterns introduced by the pipeline itself (e.g., those from selecting $\mathcal{S}$ across dimensions). As Figure 4 (middle) shows, the resulting distribution matches the centered Gaussian shape of the empirical null from independently initialized models, confirming that the Gaussian surrogate is an effective simulation-based approximation of the true null.

B.3 ANALYTICAL APPROXIMATION UNDER WEAK DEPENDENCE

Although the Gaussian surrogate captures the null behavior under the full pipeline, it still requires repeated sampling and injects randomness into the estimate. We therefore propose an even more efficient analytical approximation—the one used in the main paper.

Under the null hypothesis that the two models are independent², each Kendall–Tau correlation τ_j is approximately zero-mean with known variance σ^2 . If correlations across identity dimensions were independent, the central limit theorem would imply

$$\bar{\tau} = \frac{1}{|\mathcal{S}|} \sum_{j \in \mathcal{S}} \tau_j \sim \mathcal{N}\left(0, \frac{\sigma^2}{|\mathcal{S}|}\right).$$

In practice, softmax normalization introduces dependencies across dimensions. Yet Figure 4 shows the analytical Gaussian closely matching the Gaussian-surrogate pipeline across settings, including varying numbers of random inputs n and output dimensions D . These dependencies are thus weak enough not to materially affect the approximation; a plausible explanation is that the relatively high temperature ($T = 10$) smooths the output distribution and reduces inter-dimensional coupling.

Together, these results justify the analytical Gaussian null of Section 4 as an efficient, deterministic, and accurate approximation that eliminates costly empirical baseline construction—directly enabling the threshold-free test at the heart of Contribution 2.

²As discussed in Section B.2, although outputs may exhibit weak dependencies due to shared inputs, this approximation remains accurate empirically.

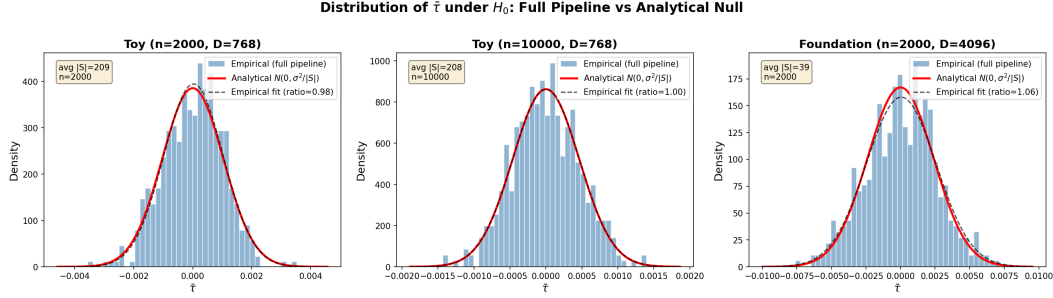


Figure 4: **The analytical null is accurate, so the test needs no empirical baseline.** We apply the full pipeline (identity-dimension selection and correlation aggregation) to Gaussian surrogate outputs and compare the resulting distribution of $\bar{\tau}$ with the analytical Gaussian $\mathcal{N}(0, \sigma^2/|S|)$. The close match shows that the analytical approximation captures the null behavior accurately despite weak pipeline-induced dependencies.

On variance scaling. The true null obtained from randomly initialized models has slightly smaller variance than the Gaussian null approximation. This mismatch does not affect our hypothesis test results. Our primary evaluation for simulation-based null in Section D.2 relies on the Mann–Whitney U-test, which is rank-based and thus invariant to any positive scaling: for any $c > 0$, $x_i < x_j \Leftrightarrow cx_i < cx_j$. Therefore, all rank statistics and rejection thresholds remain unchanged. For tests that do depend on absolute variance, we have additionally evaluated variance-sensitive statistics such as the t -test, whose statistic scales as $t' = t/c$ when variance decreases. Since $c < 1$ in our case, scaling reduces the magnitude of t and makes the test more conservative rather than inflating significance. Empirically, across extensive experiments in Section D.2, we find that the variance scaling does not affect the power of the t -test, indicating that the distributional separation between the alternative hypothesis and the null is substantially larger than the scaled variance.

For the analytical null used in the main paper, the variance is explicitly included in the normalization of the z -score. Therefore, any scale discrepancy is absorbed in the denominator. Since the analytical variance is slightly larger than the empirical one, the resulting z -scores are smaller, making the test conservative and preventing false positives.

C THEORETICAL PROPERTIES OF OUR METHODS

This section supplies the theoretical backbone of Contribution 2: it explains *why* the disfavored dimensions form a stable fingerprint signal. Our goal is to identify the output dimensions that the initialized model is least likely to predict under random inputs. These dimensions correspond to directions the model inherently disfavors due to initialization-induced bias (independent of the input distribution), and they anchor a stable fingerprint.

Setup. Consider a model g with fixed parameters and random inputs x_i (uniform random tokens or random embeddings as in the main paper). For each output dimension $j \in [d_{\text{out}}]$, $g_j(x)$ is thus a random variable and its population mean is $\bar{g}_j := \mathbb{E}[g_j(x)]$. The population-level directions we aim to capture correspond to the m output dimensions with the smallest expected values, formally defined as

$$\mathcal{M} = \arg \min_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} \bar{g}_j.$$

Let $(\bar{g}_{(1)}, \dots, \bar{g}_{(d_{\text{out}})})$ denote the sorted means in non-decreasing order, and write

$$\Gamma(m) := \bar{g}_{(m+1)} - \bar{g}_{(m)} > 0$$

for the population “gap” between the m -th and $(m+1)$ -th smallest means. This gap measures how well-separated the target set $\mathcal{M}$ is from the remaining coordinates. Importantly, for a fixed m and input distribution, $\Gamma(m)$ depends solely on the model’s inherent initialization bias. *However, increasing the length of random input sequences amplifies this bias via concentration, making the separation more pronounced at the population level.*

There are two natural empirical strategies to extract such population-level least-likely dimensions:

1. **Sample-mean-based extraction (ours).** Estimate the expectation of each output coordinate by the empirical mean over random inputs $X = \{x_i\}_{i=1}^n$ and select the m coordinates with the smallest value. Following the notation in Section 4, where g denotes the model (either the base model f or the suspect model f'):

$$\hat{g} = \frac{1}{n} \sum_{i=1}^n g(x_i), \quad \widehat{\mathcal{M}}^{\text{mean}} = \arg \min_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} \hat{g}_j.$$

2. **Per-sample argmin voting.** For each random input $x_i \in \mathbb{R}^{\ell \times d}$, select the smallest-value coordinate and then take the m most frequent dimensions:

$$j^*(x_i) = \arg \min_{j \in \{1, \dots, d_{\text{out}}\}} g_j(x_i), \quad \widehat{\mathcal{M}}^{\text{vote}} = \arg \max_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{i=1}^n \mathbf{1}\{j^*(x_i) \in J\}.$$

Let $p_j := \Pr(j^*(x) = j)$ denote the population probability that index j attains the minimum, and let

$$\mathcal{M}_{\text{vote}} := \arg \max_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} p_j$$

be the population top- m set under voting.

Both approaches are intuitively reasonable, but they are not equally reliable—and this gap is exactly why we adopt the mean-based estimator. As we prove below, the sample-mean-based strategy is provably stable, whereas per-sample argmin voting is discontinuous and can flip under arbitrarily small perturbations, making it inherently noise-sensitive.

C.1 STABILITY ANALYSIS OF MEAN-BASED IDENTITY EXTRACTION VS. PER-SAMPLE ARGMIN VOTING

C.1.1 STABILITY OF SAMPLE-MEAN-BASED EXTRACTION

We first establish that the sample-mean-based operator—which selects the m smallest coordinates of the empirical mean vector—is stable under small perturbations to model outputs. These perturbations model realistic variation in outputs from the *same* model, e.g., numerical noise or evaluation under different precision modes, exactly the conditions a deployable fingerprint must tolerate.

Formally, we model perturbed outputs as $g'(x) = g(x) + \eta(x)$, where $\eta(x) \in \mathbb{R}^{d_{\text{out}}}$ represents stochastic deviations. We assume that for each sample x_i and coordinate j , the random variables $\eta_j(x_i)$ are independent across samples (not necessarily across coordinates) and σ -sub-Gaussian with $\mathbb{E}[\eta_j(x_i)] = 0$.³

Lemma C.1 (Stability of sample-mean-based extraction). *Let $\hat{g}$ and $\hat{g}'$ be the empirical means over $X = \{x_i\}_{i=1}^n$, i.e., $\hat{g} = \frac{1}{n} \sum_{i=1}^n g(x_i)$ and $\hat{g}' = \frac{1}{n} \sum_{i=1}^n g'(x_i) = \hat{g} + \frac{1}{n} \sum_{i=1}^n \eta(x_i)$. Let $\widehat{\Gamma}(m)$ denote the empirical gap at rank m . Then,*

$$\Pr(\widehat{\mathcal{M}}^{\text{mean}}(g') \neq \widehat{\mathcal{M}}^{\text{mean}}(g)) \leq 2d_{\text{out}} \exp\left(-\frac{n\widehat{\Gamma}(m)^2}{8\sigma^2}\right).$$

Interpretation. The lemma provides a quantitative upper bound on the probability that the sample-mean-based extractor changes its selected set under perturbations. This instability probability decreases exponentially with the number of averaged samples n , due to variance reduction from σ^2 per sample to σ^2/n in the empirical mean.

Proof. We condition on the sample $X = \{x_i\}_{i=1}^n$ throughout the proof, so that $\hat{g}$ and $\widehat{\Gamma}(m)$ are fixed, and all probabilities are taken over the randomness of $\{\eta(x_i)\}_{i=1}^n$ only.

³This sub-Gaussian assumption is mild: bounded random variables are σ -sub-Gaussian (Hoeffding’s lemma), and common initialization schemes together with bounded activations yield light-tailed behavior in both initialized and trained networks. The Gaussian case $\eta_j(x_i) \sim \mathcal{N}(0, \sigma^2)$ is a special case of this assumption.

For any coordinate j , by definition of $\hat{g}'_j$, $\hat{g}'_j - \hat{g}_j = \frac{1}{n} \sum_{i=1}^n \eta_j(x_i)$. Each $\eta_j(x_i)$ is mean-zero and σ -sub-Gaussian, so their average is mean-zero and $(\sigma/\sqrt{n})$ -sub-Gaussian: the variance proxy scales as $1/n$ under averaging. Hence, for any $\varepsilon > 0$, by the standard sub-Gaussian tail bound,

$$\Pr(|\hat{g}'_j - \hat{g}_j| \geq \varepsilon \mid X) \leq 2 \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right). \quad (1)$$

Now consider the event that *any* coordinate fluctuates by more than ε :

$$\left\{ \max_{1 \leq j \leq d_{\text{out}}} |\hat{g}'_j - \hat{g}_j| \geq \varepsilon \right\}.$$

Taking a union bound over all d_{out} coordinates yields

$$\Pr\left(\max_j |\hat{g}'_j - \hat{g}_j| \geq \varepsilon \mid X\right) \leq \sum_{j=1}^{d_{\text{out}}} \Pr(|\hat{g}'_j - \hat{g}_j| \geq \varepsilon \mid X) \leq 2d_{\text{out}} \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right). \quad (2)$$

We now show that

$$\left\{ \widehat{\mathcal{M}}^{\text{mean}}(g') \neq \widehat{\mathcal{M}}^{\text{mean}}(g) \right\} \subseteq \left\{ \max_j |\hat{g}'_j - \hat{g}_j| \geq \widehat{\Gamma}(m)/2 \right\}.$$

Equivalently, if $\max_j |\hat{g}'_j - \hat{g}_j| < \widehat{\Gamma}(m)/2$, then the bottom- m identity set remains unchanged.

Let $j_{(1)}, \dots, j_{(d_{\text{out}})}$ be a permutation of $\{1, \dots, d_{\text{out}}\}$ such that $\hat{g}_{j_{(1)}} \leq \hat{g}_{j_{(2)}} \leq \dots \leq \hat{g}_{j_{(d_{\text{out}})}}$. Recall that the empirical gap at rank m is $\widehat{\Gamma}(m) := \hat{g}_{j_{(m+1)}} - \hat{g}_{j_{(m)}} > 0$. Denote $\widehat{\mathcal{M}}^{\text{mean}}(g) := \{j_{(1)}, \dots, j_{(m)}\}$ and $\widehat{\mathcal{M}}_c^{\text{mean}}(g) := \{j_{(m+1)}, \dots, j_{(d_{\text{out}})}\}$. For any $i \in \widehat{\mathcal{M}}^{\text{mean}}(g)$ and any $\ell \in \widehat{\mathcal{M}}_c^{\text{mean}}(g)$, $\hat{g}_\ell - \hat{g}_i \geq \hat{g}_{j_{(m+1)}} - \hat{g}_{j_{(m)}} = \widehat{\Gamma}(m)$. With $|\hat{g}'_i - \hat{g}_i| < \widehat{\Gamma}(m)/2$ and $|\hat{g}'_\ell - \hat{g}_\ell| < \widehat{\Gamma}(m)/2$, we have

$$\hat{g}'_\ell - \hat{g}'_i \geq (\hat{g}_\ell - \frac{\widehat{\Gamma}(m)}{2}) - (\hat{g}_i + \frac{\widehat{\Gamma}(m)}{2}) \geq \widehat{\Gamma}(m) - \widehat{\Gamma}(m) = 0.$$

Thus no coordinate in $\widehat{\mathcal{M}}_c^{\text{mean}}(g)$ can become smaller than any coordinate in $\widehat{\mathcal{M}}^{\text{mean}}(g)$ under this deviation bound, and therefore

$$\widehat{\mathcal{M}}^{\text{mean}}(g') = \widehat{\mathcal{M}}^{\text{mean}}(g).$$

Since $\widehat{\Gamma}(m)$ is fixed under conditioning on X , plugging $\varepsilon = \widehat{\Gamma}(m)/2$ into (2) yields

$$\Pr\left(\widehat{\mathcal{M}}^{\text{mean}}(g') \neq \widehat{\mathcal{M}}^{\text{mean}}(g)\right) \leq \Pr\left(\max_j |\hat{g}'_j - \hat{g}_j| \geq \widehat{\Gamma}(m)/2 \mid X\right) \leq 2d_{\text{out}} \exp\left(-\frac{n\widehat{\Gamma}(m)^2}{8\sigma^2}\right).$$

□

C.1.2 INSTABILITY OF PER-SAMPLE ARGMIN VOTING

In contrast, per-sample voting applies a discontinuous winner-take-all operator on each input. Thus, arbitrarily small perturbations to outputs can change decisions with non-vanishing probability. We still consider the same stochastic perturbation model $g'(x) = g(x) + \eta(x)$.

Lemma C.2 (Instability of per-sample argmin voting). *For a fixed input x , let*

$$j^*(x; g) := \arg \min_j g_j(x), \quad \Delta_g(x) := \min_{j \neq j^*(x; g)} \{g_j(x) - g_{j^*(x; g)}(x)\} > 0$$

denote the argmin and its margin. Then, for any fixed x ,

$$\Pr(j^*(x; g') \neq j^*(x; g) \mid x) \geq \Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right),$$

and for n i.i.d. inputs $x_1, \dots, x_n$,

$$\Pr\left(\exists i \leq n : j^*(x_i; g') \neq j^*(x_i; g)\right) \geq 1 - (1 - \alpha)^n,$$

where $\alpha := \mathbb{E}_x \left[\Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right) \right]$.

Interpretation. Unlike mean extraction, voting does not benefit from variance reduction: each input incurs a direct hard decision based on noisy outputs. Increasing n only increases the number of opportunities for index-flips.

Proof. We first bound the per-input change probability $\Pr(j^*(x; g') \neq j^*(x; g) \mid x)$. Fix x and write $j^* := j^*(x; g)$ for brevity. By definition of the margin, $\Delta_g(x) = \min_{j \neq j^*} \{g_j(x) - g_{j^*}(x)\}$. Let $j^{(2)}$ be an index attaining this minimum, i.e., $g_{j^{(2)}}(x) - g_{j^*}(x) = \Delta_g(x)$.

Under the perturbation $g'(x) = g(x) + \eta(x)$, we have

$$g'_{j^{(2)}}(x) - g'_{j^*}(x) = [g_{j^{(2)}}(x) - g_{j^*}(x)] + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)] = \Delta_g(x) + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)].$$

Since $\eta_{j^{(2)}}(x)$ and $\eta_{j^*}(x)$ are independent $\mathcal{N}(0, \sigma^2)$ variables, their difference is Gaussian with

$$\eta_{j^{(2)}}(x) - \eta_{j^*}(x) \sim \mathcal{N}(0, 2\sigma^2).$$

The event that the argmin changes, $j^*(x; g') \neq j^*(x; g)$, certainly occurs whenever $g'_{j^{(2)}}(x) \leq g'_{j^*}(x)$. Hence,

$$\Pr(j^*(x; g') \neq j^*(x; g) \mid x) \geq \Pr(g'_{j^{(2)}}(x) \leq g'_{j^*}(x) \mid x).$$

Using the expression above,

$$\Pr(g'_{j^{(2)}}(x) \leq g'_{j^*}(x) \mid x) = \Pr(g'_{j^{(2)}}(x) - g'_{j^*}(x) \leq 0 \mid x) = \Pr(\Delta_g(x) + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)] \leq 0 \mid x).$$

Let $Z \sim \mathcal{N}(0, 1)$. Since $\eta_{j^{(2)}}(x) - \eta_{j^*}(x) \stackrel{d}{=} \sqrt{2}\sigma Z$, we obtain

$$\Pr(\Delta_g(x) + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)] \leq 0 \mid x) = \Pr\left(Z \leq -\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right) = \Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right).$$

Therefore, $\Pr(j^*(x; g') \neq j^*(x; g) \mid x) \geq \Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right)$.

Now consider n i.i.d. inputs $x_1, \dots, x_n$. Conditional on the inputs, the events

$$E_i := \{j^*(x_i; g') \neq j^*(x_i; g)\} \quad E_i^c := \{j^*(x_i; g') = j^*(x_i; g)\}$$

are independent across i , because the perturbations $\eta(x_i)$ are independent across i . Moreover,

$$\Pr(E_i \mid x_i) = \Pr(j^*(x_i; g') \neq j^*(x_i; g) \mid x) \geq \Phi\left(-\frac{\Delta_g(x_i)}{\sqrt{2}\sigma}\right) \geq \Phi\left(-\frac{\Delta_g(x_i)}{\sqrt{2}\sigma}\right).$$

Define $q(x_i) := \Pr(j^*(x_i; g') \neq j^*(x_i; g) \mid x)$, and $\alpha := \mathbb{E}_x\left[\Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right)\right]$. Taking expectations over the randomness of x_i , we obtain $\mathbb{E}[\Pr(E_i \mid x_i)] = \alpha$. Therefore, by independence across i ,

$$\Pr(E_1^c \cap \dots \cap E_n^c) = \mathbb{E}\left[\prod_{i=1}^n (1 - q(x_i))\right] \leq \mathbb{E}\left[\prod_{i=1}^n (1 - \Phi(-\Delta_g(x_i)/(\sqrt{2}\sigma)))\right] \leq (1 - \alpha)^n,$$

where the last inequality uses i.i.d. inputs and Jensen's inequality applied to the convex function $u \mapsto \log(1 - u)$ on $u \in [0, 1)$. Therefore,

$$\Pr(\exists i \leq n : j^*(x_i; g') \neq j^*(x_i; g)) = 1 - \Pr(E_1^c \cap \dots \cap E_n^c) \geq 1 - (1 - \alpha)^n.$$

Finally, note that the empirical voting set $\widehat{\mathcal{M}}^{\text{vote}}$ is determined by the empirical counts of the dimensions $\{j^*(x_i; g)\}_{i=1}^n$. If the empirical gap between the m -th and $(m+1)$ -th most frequent dimensions is of order $O(1/n)$ (i.e., only a constant number of votes), then a change in $j^*(x_i)$ for a single input x_i is sufficient to swap the ordering across this boundary and thus alter the top- m set. Combining this observation with the lower bound $1 - (1 - \alpha)^n$ shows that, in such regimes, the probability that $\widehat{\mathcal{M}}^{\text{vote}}(g') \neq \widehat{\mathcal{M}}^{\text{vote}}(g)$ is bounded away from zero uniformly in n , i.e., per-sample argmin voting does not enjoy variance reduction with more samples. $\square$

C.2 INFLUENCE OF HYPERPARAMETERS

We now analyze how many random inputs are required for the empirical mean estimator to recover the true bias. We reuse notation from **Setup**: each output coordinate $g_j(x)$ is a random variable under random inputs $x \sim \mathcal{D}$, with population mean $\bar{g}_j := \mathbb{E}[g_j(x)]$. $(\bar{g}_{(1)}, \dots, \bar{g}_{(d_{\text{out}})})$ represents the sorted means in non-decreasing order, and $\gamma := \bar{g}_{(m+1)} - \bar{g}_{(m)} > 0$ is the separation margin. We estimate $\bar{g}$ using n i.i.d. random inputs:

$$\hat{g}_j := \frac{1}{n} \sum_{i=1}^n g_j(x_i), \quad \hat{g} = (\hat{g}_1, \dots, \hat{g}_{d_{\text{out}}}).$$

The target dimensions set is $\mathcal{M}$ with size m and our empirical estimation gives $\widehat{\mathcal{M}}$.

Lemma C.3 (Sufficient condition for recovering the true set). *If $\|\hat{g} - \bar{g}\|_\infty \leq \varepsilon$ and $2\varepsilon < \gamma$, then the empirical bottom- m set equals the true set: $\widehat{\mathcal{M}} = \mathcal{M}$.*

Proof. For any $j \in \mathcal{M}$ and $j' \notin \mathcal{M}$, we have $\bar{g}_j \leq \bar{g}_{(m)}$ and $\bar{g}_{j'} \geq \bar{g}_{(m+1)}$. The ℓ_∞ bound gives

$$\hat{g}_j \leq \bar{g}_{(m)} + \varepsilon, \quad \hat{g}_{j'} \geq \bar{g}_{(m+1)} - \varepsilon.$$

Thus,

$$\hat{g}_j - \hat{g}_{j'} \leq -(\gamma - 2\varepsilon) < 0,$$

so no outside index can enter nor inside index leave the bottom- m set. $\square$

We next interpret this recovery condition in terms of hyperparameters. We again adopt the same sub-Gaussian assumption as in Section C.1. This assumption is mild: by Hoeffding's inequality, any bounded random variable is σ -sub-Gaussian with σ proportional to its range. In our setting, each $g_j(x)$ corresponds to a logit or final hidden activation, which are typically bounded or light-tailed in both initialized networks (due to controlled parameter initialization) and trained networks.

Corollary C.4 (Sample complexity under sub-Gaussian coordinates). *Suppose each $g_j(x)$ is σ -sub-Gaussian. Then with probability at least $1 - \delta$,*

$$\|\hat{g} - \bar{g}\|_\infty \leq \sigma \sqrt{\frac{2 \log(2d_{\text{out}}/\delta)}{n}}.$$

Consequently, if

$$n \geq \frac{8\sigma^2}{\gamma^2} \log\left(\frac{2d_{\text{out}}}{\delta}\right),$$

then $\widehat{\mathcal{M}} = \mathcal{M}$ with probability at least $1 - \delta$.

Proof. For fixed j , sub-Gaussian concentration implies

$$\Pr(|\hat{g}_j - \bar{g}_j| > \varepsilon) \leq 2 \exp(-n\varepsilon^2/2\sigma^2).$$

A union bound over all d_{out} coordinates yields

$$\Pr\left(\|\hat{g} - \bar{g}\|_\infty > \varepsilon\right) \leq \sum_{j=1}^{d_{\text{out}}} \Pr(|\hat{g}_j - \bar{g}_j| > \varepsilon) \leq 2d_{\text{out}} \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right).$$

Set the right-hand side to δ and solve for ε to obtain, with probability at least $1 - \delta$,

$$\|\hat{g} - \bar{g}\|_\infty \leq \sigma \sqrt{\frac{2 \log(2d_{\text{out}}/\delta)}{n}}.$$

Now choose

$$\varepsilon := \sigma \sqrt{\frac{2 \log(2d_{\text{out}}/\delta)}{n}}.$$

If $n \geq \frac{8\sigma^2}{\gamma^2} \log\left(\frac{2d_{\text{out}}}{\delta}\right)$ (i.e., if $2\varepsilon < \gamma$), then Lemma C.3 applies and implies $\widehat{\mathcal{M}} = \mathcal{M}$ with probability at least $1 - \delta$. $\square$

Table 9: **Uniqueness at birth, across model families (replicates Finding 1).** Fingerprint behavior under different random initializations; all cross-seed p -values exceed 0.01.

(a) LLaMA-style models					(b) Qwen-style models				
Seed Pair	Logits Output		Hidden State		Seed Pair	Logits Output		Hidden State	
	t -test	U -test	t -test	U -test		t -test	U -test	t -test	U -test
s_{42} vs. s_{2000}	0.404	0.456	0.357	0.532	s_{123} vs. s_{1000}	0.741	0.727	0.094	0.074
s_{123} vs. s_{42}	0.214	0.295	0.678	0.565	s_{1000} vs. s_{123}	0.954	0.971	0.125	0.094
s_{1000} vs. s_{123}	0.219	0.246	0.363	0.335	s_{42} vs. s_{2000}	0.273	0.360	0.451	0.529
s_{2000} vs. s_{1000}	0.282	0.291	0.434	0.481	s_{2000} vs. s_{42}	0.215	0.206	0.230	0.295

Table 10: **Persistence through training, LLaMA-style (replicates Finding 2).** Trained models keep their initialization fingerprint ($p < 0.01$), while all baselines fail to link a model to its own seed.

Model Pair	Logits Output		Hidden State		Baselines			
	t -test	u -test	t -test	u -test	Intrinsic	REEF	PCS	ICS
s_{42}^{init} vs. s_{42}^{base}	3.33e-3✓	1.02e-3✓	2.20e-8✓	6.28e-8✓	-0.021×	0.375×	0.580×	0.196×
s_{123}^{init} vs. s_{123}^{base}	2.06e-3✓	7.33e-3✓	7.09e-6✓	1.37e-5✓	0.149×	0.369×	0.581×	0.188×
s_{1000}^{init} vs. s_{1000}^{base}	2.44e-3✓	4.14e-3✓	5.58e-4✓	2.81e-3✓	-0.252×	0.381×	0.581×	0.188×
s_{2000}^{init} vs. s_{2000}^{base}	5.63e-3✓	6.76e-3✓	4.00e-10✓	1.27e-9✓	-0.337×	0.331×	0.581×	0.188×

Therefore, accurate recovery requires $n = \Omega\left(\frac{\log d_{out}}{\gamma^2}\right)$, meaning the number of random queries grows logarithmically with the output dimension, but scales inversely with the squared separation margin γ . The margin γ itself has two contributing factors:

- **Intrinsic bias strength.** Longer random input sequences amplify initialization-induced bias via concentration, increasing the population separation between coordinates.
- **Choice of m .** Different choices of m correspond to different parts of the ranked spectrum $(\bar{g}_{(1)}, \dots, \bar{g}_{(d_{out})})$, thus yielding different effective margins in practice.

The former can be controlled by the input length l , while the latter is selected empirically based on the observed gaps in the mean spectrum.

D MORE EXPERIMENT DETAILS

D.1 IMPLEMENTATION DETAILS AND LICENSES

We train all models with the Hugging Face Transformers Trainer (Wolf et al., 2020), using Accelerate (Gugger et al., 2022) for distributed runs. All open-source models are loaded from their official Hugging Face releases and used under their original licenses: Llama models under the Meta Llama Community License, and other models under Apache-2.0. All datasets are downloaded via the Hugging Face Datasets library (the library is Apache-2.0); dataset content follows each dataset’s stated license.

D.2 COMPLEMENTARY SEEDPRINTS RESULTS USING SIMULATION-BASED NULL

This section provides complementary results that reinforce the same contributions established in Section 5, now under the empirical simulation-based null of Algorithm 1, with both *logits* and *hidden state* outputs, and tested with both the one-sided t -test (t -test) and the Mann–Whitney U test (U test). The purpose is to show that the birth-to-lifecycle findings (Contribution 3) and the baseline failures (Contribution 1) do not hinge on any single null construction or test statistic.

D.2.1 BIRTH-TO-LIFECYCLE “BIOMETRIC” FINGERPRINTING

As in the main paper, we train 12-layer, 12-head LLaMA-style models (Touvron et al., 2023) with RoPE (Su et al., 2021) and Qwen-style models (Team, 2024) from scratch. Because the simulation-

Table 11: **Persistence through training, Qwen-style (replicates Finding 2).** Trained models keep the same fingerprint behavior as their initialization ($p < 0.01$).

Model Pair	Logits Output		Hidden State	
	t -test	U -test	t -test	U -test
s_{123}^{init} vs. s_{123}^{base}	2.38e-03	1.68e-03	7.36e-15	3.38e-13
s_{1000}^{init} vs. s_{1000}^{base}	1.35e-04	4.84e-05	4.41e-13	2.01e-11
s_{42}^{init} vs. s_{42}^{base}	1.39e-03	1.46e-03	1.06e-24	2.05e-19
s_{2000}^{init} vs. s_{2000}^{base}	1.80e-03	1.28e-03	4.87e-24	1.92e-20

Table 12: **Survival under domain shift (replicates the decisive test for Contribution 1).** Fingerprint persistence under continual training on diverse datasets (base model: seed 1000, corpus openwebtext); baselines misclassify true descendants once the corpus shifts to code, while Seed-Prints attributes lineage correctly. U -test refers to the Mann–Whitney U test.

(a) LLaMA-style models

Setting	Ours (logits)		Ours (hidden)		Baselines			
Continual corpus (seed)	t -test	U -test	t -test	U -test	Intrinsic	REEF	PCS	ICS
TinyStories (1000)	0✓	0✓	0✓	7.77e-89✓	1.000✓	0.759×	0.999✓	0.996✓
TinyStories (123)	1.000✓	1.00✓	0.943✓	0.902✓	0.950×	0.658×	0.332✓	0.012✓
the_stack (1000)	0✓	1.73e-287✓	0✓	3.09e-69✓	0.489×	0.557×	0.585×	0.123×
the_stack (123)	0.616✓	0.479✓	0.732✓	0.831✓	0.445✓	0.580✓	0.301✓	0.026✓

(b) Qwen-style models

Setting	Ours (logits)		Ours (hidden)		Baselines			
Continual corpus (seed)	t -test	U -test	t -test	U -test	Intrinsic	REEF	PCS	ICS
TinyStories (1000)	5.36e-09✓	1.92e-07✓	8.49e-214✓	5.09e-71✓	1.000✓	0.957✓	0.999✓	0.996✓
TinyStories (123)	0.434✓	0.433✓	0.256✓	0.065✓	0.913×	0.199✓	0.328✓	0.039✓
the_stack (1000)	0✓	4.16e-237✓	1.16e-211✓	2.30e-76✓	0.999✓	0.313×	0.995✓	0.976✓
the_stack (123)	0.999✓	0.993✓	0.610✓	0.491✓	0.916×	0.255✓	0.328✓	0.038✓

based random baseline is stochastic, we report p -values averaged over 10 independent trials at significance level $\alpha = 0.01$. One reading caveat carries over throughout: the absolute magnitude of extremely small p -values is not meaningful—once p falls below numerical and sampling noise (e.g., $< 10^{-20}$), a value like 10^{-260} is not stronger evidence than 10^{-20} ; both decisively reject the null.

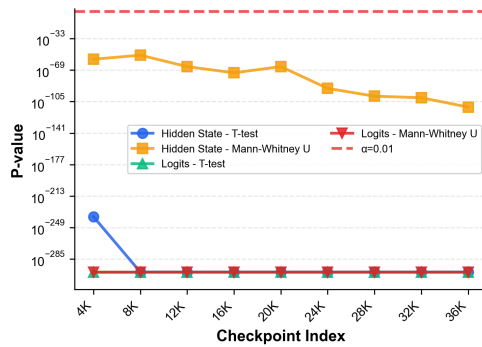
Finding 1 (replicated): different initialization seeds produce distinct fingerprints. Table 9a and Table 9b report p -values from our correlation tests between pairs of models initialized with different random seeds (42, 123, 1000, and 2000) under the t -test and U -test, for LLaMA-style and Qwen-style models respectively. All p -values are > 0.01 , so the method reliably distinguishes models trained from different seeds—distinct seeds yield distinct fingerprints, and models can be separated “at birth” regardless of the chosen null and test.

Finding 2 (replicated): training preserves the initialization fingerprint. Table 10 (LLaMA-style) and Table 11 (Qwen-style) compare each initialization model s^{init} with its descendant s^{base} trained on OpenWebText (Gokaslan et al., 2019) (≈ 10 B tokens). Every seed–model pair yields $p < 0.01$: the bias profiles stay strongly correlated and share a common lineage, so the trained model inherits its initialization fingerprint rather than losing it. Consistent with Contribution 1, the baselines in Table 10 again fail without exception to distinguish across seeds, confirming that their separability stems from training-induced artifacts rather than initialization.

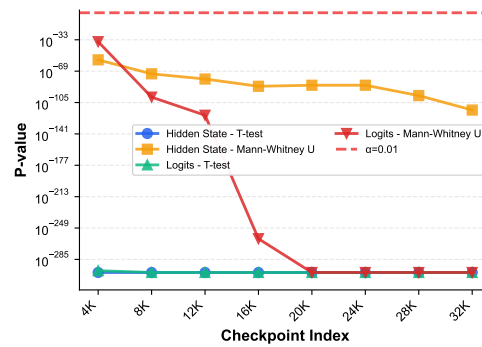
Finding 3 (replicated): identical data and order do not make fingerprints converge. In Table 13a (LLaMA-style) and Table 13b (Qwen-style), all four “suspicious” models s_i^{base} for $i \in \{42, 123, 100, 2000\}$ are trained on *exactly the same corpus (OpenWebText) and in the same*

Table 13: **Uniqueness is intrinsic, not data-driven (replicates Finding 3 under the simulation-based null)**. Identical dataset and training order do not make fingerprints converge across different initializations.

(a) LLaMA-style models						(b) Qwen-style models					
Model Pair	Logits Output		Hidden State			Model Pair	Logits Output		Hidden State		
	<i>t</i> -test	<i>U</i> -test	<i>t</i> -test	<i>U</i> -test			<i>t</i> -test	<i>U</i> -test	<i>t</i> -test	<i>U</i> -test	
s_{123}^{init} vs. s_{1000}^{base}	0.484	0.500	0.385	0.486		s_{123}^{init} vs. s_{1000}^{base}	0.598	0.638	0.286	0.254	
s_{1000}^{init} vs. s_{2000}^{base}	0.946	0.956	0.135	0.096		s_{1000}^{init} vs. s_{123}^{base}	0.804	0.805	0.236	0.240	
s_{42}^{init} vs. s_{123}^{base}	0.598	0.589	0.426	0.337		s_{42}^{init} vs. s_{2000}^{base}	0.589	0.608	0.226	0.2043	
s_{2000}^{init} vs. s_{42}^{base}	0.756	0.781	0.388	0.287		s_{2000}^{init} vs. s_{42}^{base}	0.523	0.482	0.312	0.323	



(a) LLaMA-style models



(b) Qwen-style models

Figure 5: **All-stage verifiability (Contribution 4)**. SeedPrints verifies lineage at every checkpoint (p -values < 0.01) across model families.

data order (we fix the training seed to lock the data order); only the initialization seed i differs. Testing whether identical data and order erase or confound the fingerprint, we find p -values consistently > 0.01 across all cross-seed pairs, in sharp contrast to the near-zero values in Table 10. The fingerprint therefore remains seed-specific even under identical data and curriculum.

Finding 4 (replicated): continual training on diverse datasets does not confound the fingerprint. As in the main paper, these experiments show that SeedPrints acts as a biometric fingerprint. We reiterate that prior work’s inability to distinguish seeds is not itself a copyright weakness, since initialization seeds have no clear copyright status; the genuinely damaging failure is that prior attribution can be misled by *data distribution*, *failing to recognize lineage when the training distribution shifts substantially during continued training*.

In Table 12a and Table 12b, we continue training a base model (seed 1000, pretrained on OpenWebText (Gokaslan et al., 2019)) on two very different datasets: TinyStories (Eldan & Li, 2023) (synthetic children’s stories) and The Stack (Kocetkov et al., 2022) (permissively licensed GitHub code). We compare (i) true descendants trained from the base against (ii) *distractors* derived from a different base model (initialized with seed 123, trained with a different data order on OpenWebText, then continued on the *same* corpus). The question is which descendant an attribution method identifies as truly sharing lineage with the base.

The outcome mirrors the main result: prior baselines all fail under the code setting (The Stack), misclassifying true descendants as distractors and thereby revealing that they track domain similarity rather than lineage identity—TinyStories stays close to the pretraining corpus (OpenWebText) while The Stack diverges sharply, and this large shift is enough to bypass detection. SeedPrints, by contrast, correctly attributes lineage across both corpora. Our fingerprint is thus not a proxy for data distribution: it survives substantial domain shift and persists beyond the initial pretraining stage.

Table 14: **No family-level collapse (supports Contribution 3).** Cross-size similarity between Qwen-2.5-14B (target) and other Qwen-2.5 models; SeedPrints does not flag any smaller sibling as same-lineage.

	Qwen-2.5-0.5B	Qwen-2.5-1.5B	Qwen-2.5-3B	Qwen-2.5-7B
SeedPrint ($\uparrow$)	0.7184	0.9707	0.3898	0.5429
REEF ($\downarrow$)	0.8250	0.8399	0.8833	0.8461
HUREF ($\downarrow$)	-6.37×10^{-5}	-2.84×10^{-5}	-8.70×10^{-6}	-6.08×10^{-6}

D.2.2 ALL-STAGE VERIFIABLE FINGERPRINTS

Controlled pretraining trajectory. Supporting the all-stage claim of Contribution 4, we conduct a controlled pretraining experiment on OpenWebText where the initialization seed is known. Treating each intermediate checkpoint as the base, we test whether the method identifies its offspring along the same trajectory. Every variant consistently recognizes the suspect as same-lineage, with p -values staying below the 0.01 threshold (Figure 5a for LLaMA-style and Figure 5b for Qwen-style models), confirming the stability of SeedPrints under continuous optimization.

D.3 CROSS-SIZE FINGERPRINT WITHIN THE QWEN-2.5 SERIES

A natural worry about any initialization-based fingerprint is that it might “collapse” into a single family-level signature when parameter size varies; this study shows it does not, reinforcing Contribution 3. We report cross-size similarity between Qwen-2.5-14B (target) and smaller models from the same Qwen-2.5 family. Table 14 gives results for SeedPrint ($\uparrow$) and two baselines, REEF ($\downarrow$) and HUREF ($\downarrow$).

To probe whether initialization-born fingerprints “collapse” to a family-level signature when scaling model size, we conduct a cross-size study within the Qwen-2.5 series. We treat Qwen-2.5-14B as the protected target model and consider Qwen-2.5-{0.5B, 1.5B, 3B, 7B} as suspect models that share the same architectural family but are not trained as descendants of Qwen-2.5-14B. As discussed in the main paper, SeedPrints outputs a p -value for the hypothesis test “the suspect shares initialization-born fingerprints with the target”, while REEF and HUREF are similarity-based baselines that directly output continuous scores and require the user to choose a threshold.

Three observations stand out. First, all SeedPrints p -values sit comfortably above a typical significance level (e.g., $\alpha=0.01$), so under our hypothesis-testing view none of the smaller Qwen-2.5 models is flagged as sharing lineage with Qwen-2.5-14B. This is the desired behavior: despite a shared family and architecture, the test does not collapse to a family-level signature that would spuriously conflate distinct pre-training runs.

Second, the cross-size p -values occupy different regions of $[0, 1]$ rather than concentrating at one value. We do *not* read these magnitudes as a continuous “similarity score”—their role is to support a calibrated yes/no decision at a fixed significance level—but the spread does show that our statistic stays sensitive to the specific architecture-and-seed combination rather than degenerating into a pattern common to all Qwen-2.5 variants.

Third, the similarity-based baselines behave very differently, which underscores the practical value of our threshold-free design. REEF produces scores in a narrow band (0.8250–0.8833) across all sizes, while HUREF outputs values numerically near zero (order 10^{-5}). Lacking a built-in decision rule, these methods force operators to hand-pick thresholds, and no single cutoff cleanly separates cross-size variants from genuinely lineage-related models. SeedPrints instead yields a statistical decision at a prescribed significance level, giving a clearer, more operational answer to “do these two models plausibly share a training lineage?” within a family.

E MODEL CATALOG AND DECODING SETTINGS

For full reproducibility of the Contribution 4 evaluation, this section documents the exact model pool. Our evaluation covers 58 models in total; below we record the decoding configuration used across all runs and enumerate the catalog grouped by family. For each derived model we indicate

the post-training transformation (*finetune*, *adapter*, *merge*, *quantization*, *distillation*), and “PT”/“IT” denote *pre-trained* and *instruction-tuned* source models.

Parameter	Value
max_new_tokens	512
temperature	0.7
top_p	0.9
top_k	50
do_sample	true
max_input_length	512

E.1 MODEL LIST BY FAMILY

E.1.1 QWEN-2.5-7B

Variant	HuggingFace repo	Type
PT	Qwen/Qwen2.5-7B	—
IT	Qwen/Qwen2.5-7B-Instruct	—
Derived	Qwen/Qwen2.5-Math-7B	finetune
Derived	Qwen/Qwen2.5-Coder-7B-Instruct	finetune
Derived	WangCa/Qwen2.5-7B-Medicine	finetune
Derived	huihui-ai/Qwen2.5-7B-Instruct-abliterated-v2	finetune
Derived	Locutusque/StockQwen-2.5-7B	merge
Derived	bunnycore/QevaCoT-7B-Stock	merge
Derived	fangcaotank/task-10-Qwen-Qwen2.5-7B-Instruct	adapter
Derived	SeeFlock/task-12-Qwen-Qwen2.5-7B-Instruct	adapter
Derived	Qwen/Qwen2.5-7B-Instruct-GPTQ-Int4	quantization
Derived	Qwen/Qwen2.5-7B-Instruct-GPTQ-Int8	quantization
Derived	Lansechen/Qwen2.5-7B-Open-R1-Distill	distillation

E.1.2 QWEN2.5-14B

Variant	HuggingFace repo	Type
PT	Qwen/Qwen2.5-14B	—
IT	Qwen/Qwen2.5-14B-Instruct	—
Derived	Qwen/Qwen2.5-Coder-14B	finetune
Derived	oxyapi/oxy-1-small	finetune
Derived	v000000/Qwen2.5-14B-Gutenberg-Instruct-Slerpeno	merge
Derived	ToastyPigeon/qwen-story-test-qlora	adapter
Derived	Qwen/Qwen2.5-14B-Instruct-GPTQ-Int4	quantization
Derived	deepseek-ai/DeepSeek-R1-Distill-Qwen-14B	distillation

E.1.3 LLAMA-3.1-8B

Variant	HuggingFace repo	Type
PT	meta-llama/Llama-3.1-8B	—
IT	meta-llama/Llama-3.1-8B-Instruct	—
Derived	ValiantLabs/Llama3.1-8B-Fireplace2	finetune
Derived	RedHatAI/Llama-3.1-8B-tldr	finetune
Derived	proxectonos/Llama-3.1-Carballo	finetune
Derived	mlabonne/Meta-Llama-3.1-8B-Instruct-abliterated	finetune
Derived	gaverfraxz/Meta-Llama-3.1-8B-Instruct-HalfAbliterated-TIES	merge
Derived	Xiaojian9992024/Llama3.1-8B-ExtraMix	merge
Derived	LlamaFactoryAI/Llama-3.1-8B-Instruct-cv-job-description-matching	adapter
Derived	chchen/Llama-3.1-8B-Instruct-PsyCourse-fold7	adapter
Derived	iqbalamo93/Meta-Llama-3.1-8B-Instruct-GPTQ-Q_8	quantization
Derived	DaraV/LLaMA-3.1-8B-Instruct-INT4-GPTQ	quantization
Derived	asas-ai/Llama-3.1-8B-Instruct-Open-R1-Distill	distillation

E.1.4 MISTRAL-7B-v0.3

Variant	HuggingFace repo	Type
PT	mistralai/Mistral-7B-v0.3	—
IT	mistralai/Mistral-7B-Instruct-v0.3	—
Derived	KurmaAI/AQUA-7B	finetune
Derived	openfoodfacts/spellcheck-mistral-7b	finetune
Derived	grimjim/Mistral-7B-Instruct-demi-merge-v0.3-7B	merge
Derived	chaymaemrhrioui/mistral-Brain_Model_ACC_Trainer	adapter
Derived	RedHatAI/Mistral-7B-Instruct-v0.3-GPTQ-4bit	quantization
Derived	eganwo/mistral7b-distilled-from-deepseek-r1-qwen32b	distillation

E.1.5 GEMMA-2-2B

Variant	HuggingFace repo	Type
PT	google/gemma-2-2b	—
IT	google/gemma-2-2b-it	—
Derived	rinna/gemma-2-baku-2b	finetune
Derived	anakin87/gemma-2-2b-neogenesis-ita	finetune
Derived	vonjack/gemma2-2b-merged	merge
Derived	google-cloud-partnership/gemma-2-2b-it-lora-sql	adapter
Derived	qilowoq/gemma-2-2B-it-4Bit-GPTQ	quantization
Derived	Syed-Hasan-8503/Gemma-2-2b-it-distilled	distillation

E.1.6 LLAMA-2-7B

Variant	HuggingFace repo	Type
PT	meta-llama/Llama-2-7b-hf	—
IT	meta-llama/Llama-2-7b-chat-hf	—
Derived	allenai/tulu-2-7b	finetune
Derived	QIAIUNCC/EYE-Llama_qa	finetune
Derived	DevQuasar/coma-7B-v0.1	merge
Derived	Ammar-1/llama2-Better-Tune	adapter
Derived	TheBloke/Llama-2-7B-Chat-GPTQ	quantization
Derived	cygu/llama-2-7b-logit-watermark-distill-kgw-k1-gamma0.25-delta2	distillation

E.1.7 TINYLLAMA-1.1B

Variant	HuggingFace repo	Type
PT/IT	TinyLlama/TinyLlama-1.1B-Chat-v1.0	—
Derived	alexredna/TinyLlama-1.1B-Chat-v1.0-reasoning-v2	finetune
Derived	Edentns/DataVortexTL-1.1B-v0.1	finetune
Derived	appvoid/dot-v2.7	merge
Derived	barissglc/tinyllama-tarot-v1	adapter
Derived	TheBloke/TinyLlama-1.1B-Chat-v1.0-GPTQ	quantization
Derived	anudaw/distilled-code-llama	distillation

Notes. “Type” denotes the post-training transformation relative to the PT/IT source model, spanning the full range of alterations our robustness contribution is tested against: *finetune* covers supervised/preference-optimization variants; *adapter* covers LoRA/QLoRA-style modules; *merge* covers model soups and TIES-style merges; *quantization* covers GPTQ/INTx variants; and *distillation* covers student models distilled from reasoning-augmented teachers.

E.2 SAMPLE RESULTS

To make the deployment behavior behind Contribution 4 concrete, we present partial results from one LeafBench evaluation run in Table 15 (*u*-test with simulation-based null). Reported metrics are $1 - p$, used as similarity scores. We adopt a significance level of 0.01 for *p*-values; empirically, LeafBench identifies an optimal decision threshold of 0.9920, further supporting the reliability of our method.

The key point is one of interpretation: while individual *p*-values should not be read as linear similarity measures, their distribution across model pairs is highly informative—same-lineage models yield much lower *p*-values, whereas unrelated models remain near uniform.

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Model	Qwen-2.5-7B	Qwen-2.5-7B-Instruct	Qwen2.5-14B	Qwen2.5-14B-Instruct
Qwen-2.5-7B	1.000	1.000	0.079	0.000
Qwen-2.5-7B-Instruct	1.000	1.000	0.813	0.536
Qwen2.5-7B-Math	1.000	0.994	0.047	0.987
Qwen2.5-7B-Coder	0.996	0.999	0.177	0.002
Qwen2.5-7B-Instruct-Medicine	1.000	1.000	0.576	0.369
Qwen2.5-7B-Instruct-Abiliterated	1.000	1.000	0.276	0.886
Qwen2.5-7B-Stock	1.000	1.000	0.939	0.225
QevaCoT-7B	1.000	1.000	0.199	0.119
Qwen2.5-7B-Instruct-Task-10	1.000	1.000	0.672	0.090
Qwen2.5-7B-Instruct-Task-12	1.000	1.000	0.155	0.987
Qwen2.5-7B-Instruct-Int4	1.000	1.000	0.827	0.961
Qwen2.5-7B-Instruct-Int8	1.000	1.000	0.914	0.739
Qwen2.5-7B-Open-R1-Distill	1.000	1.000	0.302	0.955
Qwen2.5-14B	0.324	0.487	1.000	1.000
Qwen2.5-14B-Instruct	0.403	0.702	1.000	1.000
Qwen2.5-Coder-14B	0.492	0.660	1.000	1.000
oxy-1-small	0.492	0.104	1.000	1.000
Qwen2.5-14B-Gutenberg-Instruct-Slerpeno	0.856	0.931	1.000	1.000
Qwen-story-test-qlora	0.112	0.898	1.000	1.000
Qwen2.5-14B-Instruct-GPTQ-Int4	0.583	0.092	1.000	1.000
DeepSeek-R1-Distill-Qwen-14B	0.204	0.000	1.000	1.000
Llama-3.1-8B	0.638	0.714	0.156	0.000
Llama-3.1-8B-Instruct	0.643	0.118	0.128	0.510
Llama-3.1-8B-Fireplace2	0.715	0.532	0.065	0.354
Llama-3.1-8B-TLDR	0.925	0.811	0.331	0.014
Llama-3.1-8B-Carballo	0.001	0.605	0.003	0.946
Llama-3.1-8B-Instruct-Abiliterated	0.061	0.005	0.121	0.422
Llama-3.1-8B-Instruct-HalfAbiliterated-TIES	0.356	0.509	0.796	0.000
Llama-3.1-8B-ExtraMix	0.640	0.160	0.721	0.875
Llama-3.1-8B-Instruct-cv-job-description-matching	0.684	0.292	0.004	0.001
Llama-3.1-8B-Instruct-PsyCourse-fold7	0.682	0.057	0.063	0.191
Llama-3.1-8B-Instruct-8bit	0.798	0.214	0.154	0.385
Llama-3.1-8B-Instruct-4bit	0.513	0.023	0.005	0.933
Llama-3.1-8B-Instruct-Open-R1-Distill	0.800	0.101	0.772	0.024
Mistral-7B-v0.3	0.224	0.352	0.506	0.044
Mistral-7B-v0.3-Instruct	0.000	0.000	0.984	0.104
AQUA-7B	0.014	0.613	0.703	0.569
Mistral-7B-v0.3-Spellcheck	0.005	0.481	0.872	0.105
Mistral-7B-v0.3-Instruct-demi-merge	0.904	0.774	0.058	0.522
Mistral-7B-v0.3-Brain	0.340	0.983	0.481	0.016
Mistral-7B-v0.3-Instruct-GPTQ-4bit	0.002	0.281	0.089	0.784
Mistral-7B-distilled-from-deepseek-r1-qwen32b	0.009	0.001	0.511	0.015
Gemma-2-2b	0.747	0.869	0.689	0.000
Gemma-2-2b-it	0.944	0.039	0.626	0.060
Gemma-2-baku-2b	0.616	0.231	0.176	0.464
Gemma-2-2b-neogenesis-ita	0.002	0.536	0.079	0.980
Gemma-2-2b-merged	0.951	0.012	0.651	0.075
Gemma-2-2b-it-lora-sql	0.236	0.408	0.000	0.000
Gemma-2-2B-it-4Bit-GPTQ	0.945	0.159	0.000	0.008
Gemma-2-2b-it-distilled	0.482	0.573	0.271	0.276
Llama-2-7b	0.443	0.470	0.148	0.276
Llama-2-7b-chat	0.528	0.868	0.424	0.766
tulu-2-7b	0.125	0.272	0.170	0.888
EYE-Llama_qa	0.657	0.244	0.674	0.968
coma-7B-v0.1	0.953	0.004	0.573	0.731
llama2-Better-Tune	0.136	0.699	0.469	0.860
Llama-2-7B-Chat-GPTQ	0.594	0.603	0.582	0.424
llama-2-7b-logit-watermark-distill-kgw-k1-gamma0.25-delta2	0.349	0.224	0.366	0.976

Table 15: **Sample LeafBench results illustrating Contribution 4.** Pairwise similarity between target models (rows) and base models (columns); same-lineage pairs stand out clearly. Cells with $1 - p \geq 0.99$ are highlighted in green as same-lineage.